

Solution Manual of QFT Assignments (Problem Set 22–35)

Bufan Zheng
bufan@g.ecc.u-tokyo.ac.jp

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Solution Manual of Problem Set 22

Bufan Zheng
bufan@g.ecc.u-tokyo.ac.jp

Conventions: mostly-plus metric $\eta^{\mu\nu} = \text{diag}(-, +, \dots, +)$ and $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$.

1. Old vs modern normalization (field redefinition). Start from

$$\mathcal{L}_{\text{old}} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \bar{\psi}(\gamma^\mu\partial_\mu + m)\psi - e\bar{\psi}\gamma^\mu A_\mu\psi.$$

Define a rescaled gauge field $A_\mu = e A'_\mu$. Rewrite the Lagrangian in terms of A'_μ and show it becomes

$$\mathcal{L}_{\text{modern}} = -\frac{1}{4e^2}F'^{\mu\nu}F'_{\mu\nu} - \bar{\psi}(\gamma^\mu\partial_\mu + m)\psi - \bar{\psi}\gamma^\mu A'_\mu\psi.$$

Solution. (22.1) $F \sim \partial A$, so **MY FINAL ANSWER IS:**

$$\begin{aligned}\mathcal{L}_{\text{old}} &= -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu A_\mu\psi \\ &= -\frac{1}{4e^2}(eF^{\mu\nu})(eF_{\mu\nu}) + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - \bar{\psi}\gamma^\mu(eA_\mu)\psi \\ &= -\frac{1}{4e^2}F'^{\mu\nu}F'_{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - \bar{\psi}\gamma^\mu A'_\mu\psi\end{aligned}$$

Note that the Lagrangian in this problem is wrong. □

2. Gauge transformation in both normalizations. In the old normalization, take $\psi \rightarrow e^{i\alpha}\psi$, $A_\mu \rightarrow A_\mu - \partial_\mu\alpha/e$. Show that in the modern normalization $A'_\mu \rightarrow A'_\mu - \partial_\mu\alpha$. Derive the corresponding covariant derivative $D_\mu = \partial_\mu + iA'_\mu$.

Solution. (22.2)

$$A_\mu \rightarrow A_\mu - \partial_\mu\alpha/e \quad \Rightarrow \quad eA_\mu \rightarrow eA_\mu - \partial_\mu\alpha \quad \Rightarrow \quad A'_\mu \rightarrow A'_\mu - \partial_\mu\alpha$$

Focus on the modern lagrangian:

$$\begin{aligned}\mathcal{L}_{\text{modern}} &= -\frac{1}{4e^2}F'^{\mu\nu}F'_{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - \bar{\psi}\gamma^\mu A'_\mu\psi \\ &= -\frac{1}{4e^2}F'^{\mu\nu}F'_{\mu\nu} + \bar{\psi}(i\gamma^\mu(\partial_\mu + iA'_\mu) - m)\psi \\ &\Rightarrow D_\mu = \partial_\mu + iA'_\mu\end{aligned}$$

□

3. Current and coupling dependence. Define the Noether current j^μ for the global $U(1)$ symmetry. Show that the coupling e appears only in the Maxwell term in the modern normalization, and not in the current coupling $j^\mu A'_\mu$.

Solution. (22.3) Consider the global $U(1)$ symmetry:

$$\psi \rightarrow e^{i\alpha}\psi, \quad \bar{\psi} \rightarrow \bar{\psi}e^{-i\alpha}, \quad A_\mu \rightarrow A_\mu$$

The infinitesimal form is:

$$\delta\psi = i\alpha\psi, \quad \delta\bar{\psi} = -i\alpha\bar{\psi}$$

The corresponding Noether current is:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \delta \psi + \delta \bar{\psi} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar{\psi})} = (i\bar{\psi}\gamma^\mu)(i\alpha\psi) = \alpha\bar{\psi}\gamma^\mu\psi$$

we can drop out α and obtain:

$$j^\mu = \bar{\psi}\gamma^\mu\psi$$

The interacting term can be rewritten in modern normalization as:

$$\mathcal{L}_{\text{int}} = -ej^\mu A_\mu = -ej^\mu \frac{A'_\mu}{e} = \boxed{-j^\mu A'_\mu},$$

It is clear the result does not contain any e . □

4. Beta function for e vs beta function for $1/e^2$. Assume $\beta(e) = be^3$ at one loop in 4D. Compute

$$\mu \frac{d}{d\mu} \left(\frac{1}{e^2} \right)$$

and show it is proportional to a constant (independent of e) at one loop.

Solution. (22.4) Using the definition $\beta(e) := \mu \frac{de}{d\mu}$, we have:

$$\mu \frac{d}{d\mu} \left(\frac{1}{e^2} \right) = -\frac{2}{e^3} \mu \frac{de}{d\mu} = -\frac{2}{e^3} \beta(e) = -2b$$

It is clear that it is proportional to a constant b at one loop. □

5. Vacuum polarization and running in the modern normalization. Explain why in the modern normalization the renormalization of the Maxwell term can be interpreted as the running of $1/e^2(\mu)$. State the sign of the running in QED and its physical meaning.

Solution. (22.5) Consider the 1PI diagram of photon propagator, $i\Pi^{\mu\nu}(q)$. The only tensors that can appear in $\Pi^{\mu\nu}(q)$ are $g^{\mu\nu}$ and $q^\mu q^\nu$. The Ward identity, however, tells us that $q_\mu \Pi^{\mu\nu}(q) = 0$. This implies that $\Pi^{\mu\nu}(q)$ is proportional to the projector $(g^{\mu\nu} - q^\mu q^\nu / q^2)$. It is therefore convenient to extract the tensor structure from $\Pi^{\mu\nu}$ in the following way:

$$\Pi^{\mu\nu}(q) = (q^2 g^{\mu\nu} - q^\mu q^\nu) \Pi(q^2)$$

Using this notation, the exact photon two-point function is:

$$\begin{aligned} \Delta^{\mu\nu}(q) &= \frac{-ig_{\mu\nu}}{q^2} + \frac{-ig_{\mu\rho}}{q^2} \left(\delta_\nu^\rho - \frac{q^\rho q_\nu}{q^2} \right) (\Pi(q^2) + \Pi^2(q^2) + \dots) \\ &= \frac{-i}{q^2(1 - \Pi(q^2))} \left(g_{\mu\nu} - \frac{q_\mu q_\nu}{q^2} \right) + \frac{-i}{q^2} \left(\frac{q_\mu q_\nu}{q^2} \right). \end{aligned}$$

According to the Ward identity, term proportional to q will not contribute to the S-matrix, so we can conclude:

$$\Delta^{\mu\nu}(q) = \frac{-ig_{\mu\nu}}{q^2(1 - \Pi(q^2))},$$

which implies the mass of photon will never be corrected in quantum level, but the electric charge will run. Because the residue of the $q^2 = 0$ pole is

$$\frac{1}{1 - \Pi(0)} \equiv Z_3.$$

The amplitude for any low- q^2 scattering process will be shifted by this factor, relative to the tree-level approximation:

$$\dots \frac{e^2 g_{\mu\nu}}{q^2} \dots \longrightarrow \dots \frac{Z_3 e^2 g_{\mu\nu}}{q^2} \dots$$

Since a factor of e lies at each end of the photon propagator, we can conveniently account for this shift by making the replacement $e \rightarrow \sqrt{Z_3}e$. This replacement is called charge renormalization.

In the modern normalization this point is more transparent, because

$$\mathcal{L}_{\text{modern}}^{\text{ren}} \supset -\frac{Z_3}{4e^2} F'_{\mu\nu} F'^{\mu\nu} = -\frac{1}{4e(\mu)^2} F'_{\mu\nu} F'^{\mu\nu},$$

where $e(\mu) = e/\sqrt{Z_3}$. The beta function of QED at 1-loop is:

$$\beta(e) \equiv \mu \frac{de}{d\mu} = \frac{e^3}{12\pi^2} + \dots > 0$$

Therefore, the sign of the running in QED is **negative**:

$$\mu \frac{d}{d\mu} \left(\frac{1}{e^2(\mu)} \right) < 0$$

The physical meaning is that vacuum polarization polarizes the vacuum as if it were a medium, producing a screening effect. At low energies (long distances), one observes a smaller effective charge that is screened by the surrounding electron-positron cloud. When probing shorter distances (larger μ or larger q^2), one penetrates the screening more deeply and sees a more “bare” charge, so the effective coupling increases. $\square$

6. Landau pole in terms of $1/e^2$. If $1/e^2(\mu) = 1/e^2(\mu_0) - k \ln(\mu/\mu_0)$, solve for the scale Λ_{Landau} where $e(\mu)$ diverges.

Solution. (22.6) As $e^2(\Lambda_{\text{Landau}})$ goes to infinity, we have:

$$1/e^2(\mu_0) - k \ln(\Lambda_{\text{Landau}}/\mu_0) = 0$$

So, **MY FINAL ANSWER IS**

$$\Lambda_{\text{Landau}} = \mu_0 \exp \left(\frac{1}{k e^2(\mu_0)} \right)$$

$\square$

7. Dimensional analysis in $D \neq 4$. In D spacetime dimensions, determine the mass dimension of e . Show that in $D = 4 - \epsilon$ the classical contribution to $\beta(e)$ is $\beta(e) = (D - 4)e/2$.

Solution. (22.7) In D spacetime dimensions (with $\hbar = c = 1$), the action is dimensionless, so the Lagrangian density has mass dimension

$$[\mathcal{L}] = D, \quad [\partial_\mu] = 1.$$

Mass dimension of e . From the Maxwell term

$$\mathcal{L} \supset -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad F_{\mu\nu} \sim \partial_\mu A_\nu,$$

we obtain

$$[F_{\mu\nu}] = 1 + [A], \quad [F^2] = 2(1 + [A]) = D \quad \Rightarrow \quad [A] = \frac{D - 2}{2}.$$

From the Dirac kinetic term

$$\mathcal{L} \supset \bar{\psi} i \gamma^\mu \partial_\mu \psi,$$

it follows that

$$[\bar{\psi}] + 1 + [\psi] = D, \quad [\bar{\psi}] = [\psi] \quad \Rightarrow \quad [\psi] = \frac{D-1}{2}.$$

Using the interaction term

$$\mathcal{L}_{\text{int}} = -e \bar{\psi} \gamma^\mu A_\mu \psi,$$

we require

$$[e] + [\bar{\psi}] + [A] + [\psi] = D,$$

hence

$$[e] = D - 2[\psi] - [A] = D - (D-1) - \frac{D-2}{2} = \frac{4-D}{2}.$$

Therefore

$$\boxed{[e] = \frac{4-D}{2}}.$$

Classical contribution to $\beta(e)$. From the above dimensional analysis, we have:

$$e_0 = \mu^{\frac{4-D}{2}} e(\mu),$$

Since the bare coupling e_0 is μ -independent, we have

$$0 = \mu \frac{d}{d\mu} \ln e_0 = \frac{4-D}{2} + \mu \frac{d}{d\mu} \ln e(\mu),$$

which implies

$$\boxed{\beta_{\text{class}}(e) \equiv \mu \frac{de}{d\mu} = \frac{D-4}{2} e}$$

□

Solution Manual of Problem Set 23

Bufan Zheng

bufan@g.ecc.u-tokyo.ac.jp

Conventions: mostly-plus metric. Use modern gauge-field normalization $\mathcal{L} \supset -\frac{1}{4e^2}F^2$ so that $D_\mu = \partial_\mu + iA_\mu$ and $A_\mu \rightarrow A_\mu - \partial_\mu\alpha$.

1. Higgs mechanism: vector mass. Consider a complex scalar Φ of charge 1 with

$$\mathcal{L} = -\frac{1}{4e^2}F^2 + |D_\mu\Phi|^2 - V(|\Phi|^2), \quad V(|\Phi|^2) = \lambda\left(|\Phi|^2 - \frac{v^2}{2}\right)^2.$$

Expand around $\Phi = \frac{1}{\sqrt{2}}(v+h)e^{i\pi/v}$ and show the gauge boson mass is $m_A = ev$.

Solution. (23.1)

$$\begin{aligned} \mathcal{L} &= -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + |D_\mu\Phi|^2 - \lambda\left(|\Phi|^2 - \frac{v^2}{2}\right)^2 \\ &= -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + \left|(\partial_\mu + iA_\mu)\frac{1}{\sqrt{2}}(v+h)e^{i\pi/v}\right|^2 - \lambda\left(\frac{(v+h)^2}{2} - \frac{v^2}{2}\right)^2 \\ &= -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\left|\partial_\mu h + i(v+h)\left(A_\mu + \frac{\partial_\mu\pi}{v}\right)\right|^2 - \frac{\lambda}{4}((v+h)^2 - v^2)^2 \\ &= -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}(v+h)^2\left(A_\mu + \frac{\partial_\mu\pi}{v}\right)^2 - \frac{\lambda}{4}(2vh + h^2)^2 \\ &= -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}(v^2 + 2vh + h^2)\left(A_\mu A^\mu + \frac{2}{v}A^\mu\partial_\mu\pi + \frac{1}{v^2}(\partial_\mu\pi)^2\right) - \lambda\left(v^2h^2 + vh^3 + \frac{1}{4}h^4\right) \\ &= -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}v^2A_\mu A^\mu + vhA_\mu A^\mu + \frac{1}{2}h^2A_\mu A^\mu + vA^\mu\partial_\mu\pi + 2hA^\mu\partial_\mu\pi + \frac{h^2}{v}A^\mu\partial_\mu\pi \\ &\quad + \frac{1}{2}(\partial_\mu\pi)^2 + \frac{h}{v}(\partial_\mu\pi)^2 + \frac{h^2}{2v^2}(\partial_\mu\pi)^2 - \lambda v^2h^2 - \lambda vh^3 - \frac{\lambda}{4}h^4 \\ &= -\frac{1}{4}F_{\mu\nu}^{(c)}F^{(c)\mu\nu} + \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}e^2v^2A_\mu^{(c)}A^{(c)\mu} + e^2vhA_\mu^{(c)}A^{(c)\mu} + \frac{1}{2}e^2h^2A_\mu^{(c)}A^{(c)\mu} \\ &\quad + evA^{(c)\mu}\partial_\mu\pi + 2ehA^{(c)\mu}\partial_\mu\pi + \frac{eh^2}{v}A^{(c)\mu}\partial_\mu\pi + \frac{1}{2}(\partial_\mu\pi)^2 + \frac{h}{v}(\partial_\mu\pi)^2 + \frac{h^2}{2v^2}(\partial_\mu\pi)^2 - \lambda v^2h^2 - \lambda vh^3 - \frac{\lambda}{4}h^4 \\ &\supset \frac{1}{2}m_A^2A_\mu^{(c)}A^{(c)\mu} = \frac{1}{2}e^2v^2A_\mu^{(c)}A^{(c)\mu} \Rightarrow \boxed{m_A = ev} \end{aligned}$$

Where $\square^{(c)}$ means the old normalization. □

2. Unitary gauge. Perform a gauge transformation to set $\pi = 0$. Write the quadratic Lagrangian for A_μ and h and identify all masses.

Solution. (23.2) From the $U(1)$ gauge transformation,

$$\Phi(x) \rightarrow \Phi'(x) = e^{-i\alpha(x)}\Phi(x), \quad A_\mu(x) \rightarrow A'_\mu(x) = A_\mu(x) + \partial_\mu\alpha(x)$$

We can choose, $\alpha(x) = \pi(x)/v$, so that

$$\Phi' = \frac{1}{\sqrt{2}}(v+h), \quad A'_\mu = A_\mu + \frac{1}{v}\partial_\mu\pi$$

$$\begin{aligned}
\Rightarrow \mathcal{L} &= -\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} + |D_\mu \Phi|^2 - \lambda \left(|\Phi|^2 - \frac{v^2}{2} \right)^2 \\
&= -\frac{1}{4e^2} F'_{\mu\nu} F'^{\mu\nu} + \left| (\partial_\mu + iA'_\mu) \frac{1}{\sqrt{2}} (v+h) \right|^2 - \lambda \left(\frac{(v+h)^2}{2} - \frac{v^2}{2} \right)^2 \\
&= -\frac{1}{4e^2} F'_{\mu\nu} F'^{\mu\nu} + \frac{1}{2} (\partial_\mu h)^2 + \frac{1}{2} (v+h)^2 A'_\mu A'^\mu - \frac{\lambda}{4} ((v+h)^2 - v^2)^2 \\
&= -\frac{1}{4e^2} F'_{\mu\nu} F'^{\mu\nu} + \frac{1}{2} (\partial_\mu h)^2 + \frac{1}{2} v^2 A'_\mu A'^\mu + v h A'_\mu A'^\mu + \frac{1}{2} h^2 A'_\mu A'^\mu - \lambda v^2 h^2 - \lambda v h^3 - \frac{\lambda}{4} h^4 \\
&= -\frac{1}{4e^2} (\partial_\mu A'_\nu - \partial_\nu A'_\mu) (\partial^\mu A'^\nu - \partial^\nu A'^\mu) + \frac{1}{2} (\partial_\mu h)^2 + \left(\frac{1}{2} v^2 + v h + \frac{1}{2} h^2 \right) A'_\mu A'^\mu - \lambda v^2 h^2 - \lambda v h^3 - \frac{\lambda}{4} h^4 \\
&= -\frac{1}{4e^2} \left(\partial_\mu (A_\nu + \frac{1}{v} \partial_\nu \pi) - \partial_\nu (A_\mu + \frac{1}{v} \partial_\mu \pi) \right) \left(\partial^\mu (A^\nu + \frac{1}{v} \partial^\nu \pi) - \partial^\nu (A^\mu + \frac{1}{v} \partial^\mu \pi) \right) \\
&\quad + \frac{1}{2} (\partial_\mu h)^2 + \left(\frac{1}{2} v^2 + v h + \frac{1}{2} h^2 \right) \left(A_\mu + \frac{1}{v} \partial_\mu \pi \right) \left(A^\mu + \frac{1}{v} \partial^\mu \pi \right) - \lambda v^2 h^2 - \lambda v h^3 - \frac{\lambda}{4} h^4 \\
&= -\frac{1}{4e^2} \underbrace{(\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu)}_{F_{\mu\nu} F^{\mu\nu}} + \frac{1}{2} (\partial_\mu h)^2 + \frac{1}{2} v^2 A_\mu A^\mu + v A^\mu \partial_\mu \pi + \frac{1}{2} (\partial_\mu \pi)^2 \\
&\quad + v h A_\mu A^\mu + 2 h A^\mu \partial_\mu \pi + \frac{h}{v} (\partial_\mu \pi)^2 + \frac{1}{2} h^2 A_\mu A^\mu + \frac{h^2}{v} A^\mu \partial_\mu \pi + \frac{h^2}{2v^2} (\partial_\mu \pi)^2 \\
&\quad - \lambda v^2 h^2 - \lambda v h^3 - \frac{\lambda}{4} h^4 \\
&\supset \frac{1}{2} (\partial_\mu h)^2 - \lambda v^2 h^2 - \frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} v^2 A_\mu A^\mu
\end{aligned}$$

Although in the computation above we expanded the gauge field and it may look as if the final expression still contains terms involving π , note that all such terms can be absorbed into a gauge transformation of the gauge field. Therefore π does not correspond to a physical degree of freedom. Back to the old normalization, so the masses are

$$m_A^2 = e^2 v^2 \Rightarrow m_A = ev, \quad \frac{1}{2} m_h^2 h^2 = \lambda v^2 h^2 \Rightarrow m_h^2 = 2\lambda v^2 \Rightarrow m_h = \sqrt{2\lambda} v$$

□

3. R_ξ gauge fixing. Add

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} (\partial \cdot A - \xi m_A \pi)^2.$$

Show the A_μ - π mixing cancels and compute the π mass.

Solution. (23.3) From the calculations in problem 1, we have:

$$\begin{aligned}
\mathcal{L}_{\text{tot}} &= \left[-\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} (\partial_\mu h)^2 + \frac{1}{2} v^2 A_\mu A^\mu + v h A_\mu A^\mu + \frac{1}{2} h^2 A_\mu A^\mu + v A^\mu \partial_\mu \pi + 2h A^\mu \partial_\mu \pi + \frac{h^2}{v} A^\mu \partial_\mu \pi \right. \\
&\quad \left. + \frac{1}{2} (\partial_\mu \pi)^2 + \frac{h}{v} (\partial_\mu \pi)^2 + \frac{h^2}{2v^2} (\partial_\mu \pi)^2 - \lambda v^2 h^2 - \lambda v h^3 - \frac{\lambda}{4} h^4 \right] - \frac{1}{2\xi} (\partial \cdot A - \xi m_A \pi)^2 \\
&= -\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} (\partial_\mu h)^2 + \frac{1}{2} v^2 A_\mu A^\mu + v h A_\mu A^\mu + \frac{1}{2} h^2 A_\mu A^\mu + v A^\mu \partial_\mu \pi + 2h A^\mu \partial_\mu \pi + \frac{h^2}{v} A^\mu \partial_\mu \pi \\
&\quad + \frac{1}{2} (\partial_\mu \pi)^2 + \frac{h}{v} (\partial_\mu \pi)^2 + \frac{h^2}{2v^2} (\partial_\mu \pi)^2 - \lambda v^2 h^2 - \lambda v h^3 - \frac{\lambda}{4} h^4 - \frac{1}{2\xi} (\partial \cdot A)^2 + m_A \pi \partial \cdot A - \frac{\xi}{2} m_A^2 \pi^2 \\
&= -\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial \cdot A)^2 + \frac{1}{2} (\partial_\mu h)^2 + \frac{1}{2} v^2 A_\mu A^\mu + v h A_\mu A^\mu + \frac{1}{2} h^2 A_\mu A^\mu \\
&\quad + \left(v - m_A \right) A^\mu \partial_\mu \pi + 2h A^\mu \partial_\mu \pi + \frac{h^2}{v} A^\mu \partial_\mu \pi + \frac{1}{2} (\partial_\mu \pi)^2 + \frac{h}{v} (\partial_\mu \pi)^2 + \frac{h^2}{2v^2} (\partial_\mu \pi)^2 \\
&\quad - \frac{\xi}{2} m_A^2 \pi^2 - \lambda v^2 h^2 - \lambda v h^3 - \frac{\lambda}{4} h^4
\end{aligned}$$

In the computation above we used integration by parts, so the final result is understood to be equal up to a total-derivative term. Notice that the A_μ - π mixing interaction term is

$$\mathcal{L}_{\text{tot}} \supset (v - m_A) A^\mu \partial_\mu \pi = 0$$

This term vanishes because, from the result of the previous problem, in the modern normalization we have $m_A = v$. Now we look at the quadratic terms involving π .

$$\mathcal{L}_{\text{tot}} \supset \frac{1}{2} (\partial_\mu \pi)^2 - \frac{\xi}{2} m_A^2 \pi^2$$

From these we obtain the π mass as

$$m_\pi^2 = \xi m_A^2 = \xi v^2$$

□

4. Propagators. Compute the momentum-space propagator for A_μ in R_ξ gauge. Show it behaves as $1/p^2$ at large momentum for all components.

Solution. (23.4) From the result in the last problem, only the following terms are relevant for the free A_μ propagator:

$$\mathcal{L}_{\text{tot}} \supset -\frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial \cdot A)^2 + \frac{1}{2} v^2 A_\mu A^\mu \sim \frac{1}{2} A_\mu \left[\frac{1}{e^2} (\eta^{\mu\nu} \square - \partial^\mu \partial^\nu) + \frac{1}{\xi} \partial^\mu \partial^\nu + v^2 \eta^{\mu\nu} \right] A_\nu$$

Fourier transforming A_μ gives

$$\mathcal{L}_A = \frac{1}{2} A_\mu(-p) \left[\frac{1}{e^2} (-p^2 \eta^{\mu\nu} + p^\mu p^\nu) - \frac{1}{\xi} p^\mu p^\nu + v^2 \eta^{\mu\nu} \right] A_\nu(p).$$

Introduce the transverse and longitudinal projectors

$$P_T^{\mu\nu} = \eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}, \quad P_L^{\mu\nu} = \frac{p^\mu p^\nu}{p^2}, \quad P_T^2 = P_T, \quad P_L^2 = P_L, \quad P_T P_L = 0, \quad P_T + P_L = \eta,$$

then the inverse propagator decomposes as

$$iD_{\mu\nu}^{-1}(p) = \left[\frac{1}{e^2} (-p^2) + v^2 \right] P_T{}_{\mu\nu} + \left[v^2 - \frac{p^2}{\xi} \right] P_L{}_{\mu\nu}.$$

Inverting separately on the P_T and P_L subspaces (so that $P_{T/L} = P_{T/L}^{-1}$) yields the R_ξ -gauge propagator

$$D_{\mu\nu}(p) = -i \left[\frac{e^2}{p^2 - e^2 v^2} P_{T\mu\nu} + \frac{e^2 \xi}{p^2 - \xi e^2 v^2} P_{L\mu\nu} \right] = -i e^2 \left[\frac{\eta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}}{p^2 - m_A^2} + \frac{\xi \frac{p_\mu p_\nu}{p^2}}{p^2 - \xi m_A^2} \right], \quad m_A \equiv ev$$

At large momentum $p^2 \gg m_A^2$,

$$D_{\mu\nu}(p) \sim -i e^2 \left[\frac{\eta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}}{p^2} + \frac{\xi \frac{p_\mu p_\nu}{p^2}}{p^2} \right] = -i e^2 \frac{\eta_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2}}{p^2},$$

so every component scales as $1/p^2$ for $|p| \rightarrow \infty$. □

5. Goldstone equivalence (tree-level check). At energies $E \gg m_A$, argue that amplitudes with longitudinally polarized A_μ are controlled by the Goldstone mode π . Verify in a simple tree-level process of your choice.

Solution. (23.5) Consider the Glashow–Weinberg–Salam (GWS) theory and focus on the following process:

$$e^+ e^- \rightarrow W_0^- W_0^+$$

The Goldstone equivalence states that

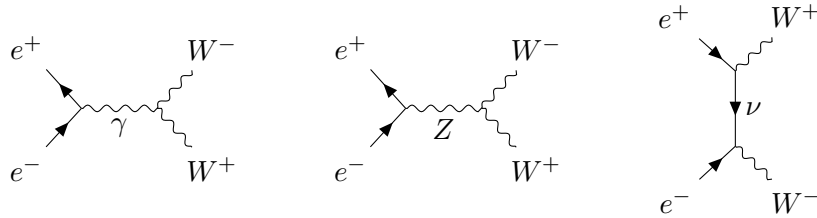
$$\mathcal{M}(e^+ e^- \rightarrow W_0^- W_0^+) = \mathcal{M}(e^+ e^- \rightarrow \pi^+ \pi^-) \times (1 + \mathcal{O}(m_W/E_W))$$

Here, W_0 means the longitudinal polarization. The RHS is easier to compute, in tree-level, we have:

$$\mathcal{M}(e_L^+ e_R^- \rightarrow \pi^+ \pi^-) = -i(2E)\sqrt{2}\epsilon_+ \cdot (k_- - k_+) \cdot \frac{e^2}{2c_w^2} \frac{1}{s}$$

$$\mathcal{M}(e_R^+ e_L^- \rightarrow \pi^+ \pi^-) = -i(2E)\sqrt{2}\epsilon_- \cdot (k_- - k_+) \cdot \left(\frac{e^2}{4c_w^2} \frac{1}{s} + \frac{e^2}{4s_w^2} \frac{1}{s} \right)$$

In fact, at leading order in the Standard Model, the LHS is the sum of 3 diagrams:



Work this out carefully for $e_R^- e_L^+$, where the ν diagram is absent:

$$i\mathcal{M} = (-ie)(ie)2E\sqrt{2}\epsilon_{+\mu} \left[\frac{-i}{s} + \frac{-s_w^2 c_w}{s_w c_w s_w} \frac{-i}{s - m_Z^2} \right] \cdot [\epsilon_-^* \epsilon_+^* (k_- - k_+)^\mu + \epsilon_-^{*\mu} (-q - k_-) \cdot \epsilon_+^* + \epsilon_+^{*\mu} (q + k_+) \cdot \epsilon_-^*]$$

where $q = k_- + k_+$ and, in the 2nd line, ϵ_-^* and ϵ_+^* are the W polarizations. Send

$$\epsilon_-^* \rightarrow \frac{k_-}{m_W}, \quad \epsilon_+^* \rightarrow \frac{k_+}{m_W}.$$

Then the second term in brackets becomes

$$\frac{1}{m_W^2} \left[2k_- \cdot k_+ (k_- - k_+)^{\mu} + k_-^{\mu} (-2k_- \cdot k_+) + k_+^{\mu} (2k_+ \cdot k_-) \right] = -\frac{k_+ \cdot k_-}{m_W^2} (k_- - k_+)^{\mu} = -\frac{s - 2m_W^2}{2m_W^2} (k_- - k_+)^{\mu}.$$

On the other hand, the first term in brackets becomes

$$\left[\frac{-i}{s} - \frac{-i}{s - m_Z^2} \right] = \frac{im_Z^2}{s(s - m_Z^2)}$$

Using $m_Z^2/m_W^2 = 1/c_w^2$, we find

$$i\mathcal{M}(e_L^+ e_R^- \rightarrow W_0^- W_0^+) = i(e^2)2E\sqrt{2}\epsilon_{+\mu}(k_- - k_+)^{\mu} \left(\frac{1}{s(s - m_Z^2)} \right) \left(-\frac{s - 2m_W^2}{2c_w^2} \right)$$

which gives the predicted expression at high energy ($s \gg m_{W/Z}^2$). For $e_R^+ e_L^-$, we must work a little harder. The first two diagrams contribute

$$i\mathcal{M} = (-ie)(ie)2E\sqrt{2}\epsilon_{-\mu} \left[\frac{-i}{s} + \frac{(1/2 - s_w^2)}{s_w c_w} \frac{c_w}{s_w} \frac{-i}{s - m_Z^2} \right] \cdot [\epsilon_-^* \epsilon_+^* (k_- - k_+)^{\mu} + \epsilon_-^* \mu(-q - k_-) \cdot \epsilon_+^* + \epsilon_+^* \mu(q + k_+) \cdot \epsilon_-^*]$$

After the reductions described above, there is a term that does not cancel its high energy behavior

$$\begin{aligned} i\mathcal{M} &\supset (-ie^2)2E\sqrt{2}\epsilon_{-\mu} \left[\frac{1}{2s_w^2} \frac{1}{s} \right] (-\frac{s}{2m_W^2} (k_- - k_+)^{\mu}) \\ &= (\frac{ie^2}{4s_w^2})2E\sqrt{2}\epsilon_{-\mu} \frac{1}{m_W^2} (k_- - k_+)^{\mu} \end{aligned}$$

However, now we must add the third diagram, which contributes

$$i\mathcal{M}_3 = (i\frac{g}{\sqrt{2}})^2 v_R^{\dagger} \bar{\sigma} \cdot \epsilon_+^* \frac{i\sigma \cdot (p - k_-)}{(p - k_-)^2} \bar{\sigma} \cdot \epsilon_-^* u_L(p)$$

Substitute $\epsilon_-^* \rightarrow k_-/m_W$ (the reason can be found in problem 24.1) and simplify

$$\frac{\sigma \cdot (p - k_-)}{(p - k_-)^2} \bar{\sigma} \cdot \frac{k_-}{m_W} u(p) = \frac{\sigma \cdot (p - k_-)}{(p - k_-)^2} \bar{\sigma} \cdot \frac{(k_- - p)}{m_W} u(p) = -\frac{1}{m_W} u(p)$$

Also $\epsilon_+^* \rightarrow k_+/m_W$ and $k_+ = (k_+ - k_- + (p + \bar{p}))/2$. So finally we find

$$i\mathcal{M}_3 = -i\frac{e^2}{2s_w^2} (2E)\sqrt{2}\epsilon_{-\mu} \frac{1}{2} (k_- - k_+)^{\mu} \frac{1}{m_W^2}$$

which indeed cancels the previous term. Thanks to [Peskin's lecture notes](#) from the MITP summer school for helping me understand this point. $\square$

6. Gauge invariance of physical masses. Explain why the physical vector mass m_A is gauge-parameter independent, while the π mass depends on ξ . Identify which poles in propagators correspond to physical particles.

7. Counting degrees of freedom. Count degrees of freedom before and after Higgsing. Explain how the Goldstone mode is removed from the spectrum in unitary gauge and how it reappears as an unphysical mode in R_{ξ} .

Solution. (23.7)

Before Higgsing. The massless gauge field A_μ has 4 components, but gauge invariance and the constraint remove 2 of them, leaving

$$\#(A_\mu \text{ massless}) = 2$$

The complex scalar Φ contains two real scalars, so

$$\#(\Phi) = 2.$$

Hence the total number of physical degrees of freedom is 4.

After Higgsing. The gauge boson acquires a mass, so a massive vector field in four dimensions has

$$\#(A_\mu \text{ massive}) = 3$$

The radial Higgs mode h is one real scalar,

$$\#(h) = 1,$$

It may seem that the Goldstone field π should still contribute one degree of freedom, but in fact, as we learned from the analysis in Problem 2, it can be completely removed by a gauge transformation. In other words, the degree of freedom carried by π is converted into the longitudinal degree of freedom of A_μ . Therefore the total number of physical degrees of freedom is 4, which matches the count before Higgsing.

How the Goldstone mode disappears in unitary gauge. Under a local $U(1)$ transformation,

$$\Phi(x) \rightarrow e^{-i\alpha(x)}\Phi(x), \quad A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu\alpha(x).$$

Choosing $\alpha(x) = \pi(x)/v$ sets $\pi' = 0$ and shifts the gauge field to

$$\Phi' = \frac{1}{\sqrt{2}}(v + h), \quad A'_\mu = A_\mu + \frac{1}{v}\partial_\mu\pi.$$

Thus the Goldstone field is removed from the spectrum: it is not an independent physical excitation, but rather provides the longitudinal polarization of the now-massive vector boson.

How it reappears as an unphysical mode in R_ξ . In R_ξ gauge one adds the gauge-fixing term

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial \cdot A - \xi m_A \pi)^2,$$

As we see in problem 3, it cancels the quadratic A - π mixing and gives the Goldstone field a gauge-dependent mass,

$$m_\pi^2 = \xi m_A^2.$$

Because this mass depends on the gauge parameter ξ , π does not correspond to a physical particle. It can propagate only as an internal (unphysical) mode, and all ξ -dependence cancels out of gauge-invariant observables. $\square$

Solution Manual of Problem Set 24

Bufan Zheng
bufan@g.ecc.u-tokyo.ac.jp

Conventions: mostly-plus metric. Use modern normalization $\mathcal{L} \supset -\frac{1}{4e^2}F^2$ and $D_\mu = \partial_\mu + iA_\mu$.

1. Longitudinal polarization vector. For a massive vector of mass m , write the longitudinal polarization vector $\epsilon_L^\mu(p)$ and show that for $E \gg m$ it behaves like $\epsilon_L^\mu(p) \sim p^\mu/m$ up to corrections of order m/E .

Solution. (24.1) In the rest frame, a massive vector boson has 3 polarization states

$$\epsilon_\pm^\mu = (1/\sqrt{2})(0, 1, \pm i, 0)^\mu, \quad \epsilon_L^\mu = (0, 0, 0, 1)^\mu$$

representing the 3 possible states of a spin 1 particle with $J^3 = \pm 1, 0$. Now boost along the 3 axis to high energy. The boosts of the polarization vectors are

$$\epsilon_\pm^\mu = (1/\sqrt{2})(0, 1, \pm i, 0)^\mu, \quad \epsilon_L^\mu = \left(\frac{p}{m}, 0, 0, \frac{E}{m}\right)^\mu$$

Note that, as E becomes large, the components of ϵ_L^μ grow without bound, in fact:

$$\epsilon_0^\mu \rightarrow \frac{p^\mu}{m}$$

□

2. Equivalence theorem statement. State the Goldstone equivalence theorem in the Abelian Higgs model. Describe the regime of validity and what is meant by “up to m/E corrections.”

Solution. (24.2) The Goldstone equivalence theorem for Abelian Higgs model is as follows:

Consider the Abelian Higgs model in an R_ξ gauge, where the gauge boson A_μ acquires a mass m_A and the would-be Goldstone mode is π . For a scattering amplitude with n external *longitudinally polarized* massive gauge bosons,

$$A_L^{a_1}(p_1), A_L^{a_2}(p_2), \dots, A_L^{a_n}(p_n),$$

the Goldstone equivalence theorem states that, in the high-energy regime

$$E_k \gg m_A \quad (k = 1, \dots, n),$$

the amplitude is equal, at leading order in m_A/E_k , to the amplitude in which each external longitudinal gauge boson is replaced by the corresponding Goldstone field:

$$\mathcal{M}(\cdots A_L(p_1) \cdots A_L(p_n) \cdots) = i^n \left(\prod_{k=1}^n C_k \right) \mathcal{M}(\cdots \pi(p_1) \cdots \pi(p_n) \cdots) + \mathcal{O}\left(\sum_{k=1}^n \frac{m_A}{E_k}\right).$$

Here E_k denotes the typical energy scale associated with the external leg p_k (e.g. $E_k \simeq |p_k^0|$ in the center-of-mass frame), and the factors C_k are external-leg normalization constants that satisfy $C_k = 1$ at tree level (and receive loop-level renormalization corrections in general).

Meaning of “up to m/E corrections.” The statement “up to m/E corrections” means that the difference between the longitudinal-gauge-boson amplitude and the Goldstone amplitude is suppressed by at least one power of m_A/E (and similarly by other relevant masses over E), originating from the fact that

$$\epsilon_L^\mu(p) = \frac{p^\mu}{m_A} + \mathcal{O}\left(\frac{m_A}{E}\right),$$

so subleading terms in the longitudinal polarization vector and other non-leading contributions are parametrically small in the high-energy limit. $\square$

3. Simple amplitude comparison. Consider a process involving one external longitudinal vector. Compute the leading high-energy behavior using $\epsilon_L^\mu \simeq p^\mu/m$ and compare with the amplitude computed using the Goldstone mode π . Show agreement at leading order.

4. Gauge-fixing dependence of unphysical poles. In R_ξ gauge, identify the pole masses of A_μ (transverse and longitudinal parts) and π . Show explicitly that the unphysical pole depends on ξ .

5. Unitary gauge vs R_ξ gauge at large momentum. Explain why the unitary-gauge propagator behaves poorly at large momentum, and how R_ξ gauge restores manifest power counting. Relate this to renormalizability in perturbation theory.

6. Gauge-invariant observables. Give two examples of gauge-invariant quantities in the Abelian Higgs model and explain why their values cannot depend on ξ .

7. Stueckelberg viewpoint. Rewrite the massive vector theory using a Stueckelberg field π . Show how R_ξ gauge corresponds to adding a particular gauge-fixing functional involving π .

Solution Manual of Problem Set 25

Bufan Zheng
bufan@ecc.u-tokyo.ac.jp

Mostly-plus metric. Use $T^{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S}{\delta g_{\mu\nu}}$ in curved space, evaluated at $g = \eta$.

1. Canonical stress tensor (free scalar). For $\mathcal{L} = -\frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - \frac{m^2}{2}\varphi^2$, compute the canonical $T_{\text{can}}^{\mu\nu}$ and verify $\partial_\mu T_{\text{can}}^{\mu\nu} = 0$ on-shell.

Solution. (25.1) The canonical momentum–energy tensor is:

$$T_{\text{can}}^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu\varphi)} \partial^\nu\varphi - \eta^{\mu\nu} \mathcal{L} = -\partial^\mu\varphi \partial^\nu\varphi - \eta^{\mu\nu} \mathcal{L} = -\partial^\mu\varphi \partial^\nu\varphi + \eta^{\mu\nu} \left(\frac{1}{2}(\partial\varphi)^2 + \frac{1}{2}m^2\varphi^2 \right)$$

The conservation law can be shown as follows:

$$\partial_\mu T_{\text{can}}^{\mu\nu} = -\partial_\mu(\partial^\mu\varphi \partial^\nu\varphi) + \partial^\nu \left(\frac{1}{2}(\partial\varphi)^2 + \frac{1}{2}m^2\varphi^2 \right) = -(\square - m^2)\varphi \partial^\nu\varphi$$

Which vanishes on-shell. □

2. Energy density and positivity. Compute T^{00} for the free scalar and express it in terms of $\dot{\varphi}$ and spatial gradients $\nabla\varphi$. Show it is nonnegative for real φ .

Solution. (25.2)

$$T^0_0 = \dot{\varphi}^2 - \mathcal{L} = \frac{1}{2}\dot{\varphi}^2 + \frac{1}{2}(\nabla\varphi)^2 + \frac{1}{2}m^2\varphi^2 \geq 0$$

□

3. Symmetrization via metric variation. Minimally couple the free scalar to $g_{\mu\nu}$. Compute $T^{\mu\nu}$ by varying $g_{\mu\nu}$ and show it is symmetric.

Solution. (25.3) The minimally coupled action is:

$$\begin{aligned} S &= \int d^Dx \sqrt{-g} \left(-\frac{1}{2}g^{\mu\nu} \nabla_\mu\varphi \nabla_\nu\varphi - \frac{1}{2}m^2\varphi^2 \right) \\ \delta S &= \delta \int d^Dx \sqrt{-g} \left(-\frac{1}{2}g^{\mu\nu} \nabla_\mu\varphi \nabla_\nu\varphi - \frac{1}{2}m^2\varphi^2 \right) \\ &= \int d^Dx \left[\delta(\sqrt{-g}) \left(-\frac{1}{2}g^{\mu\nu} \nabla_\mu\varphi \nabla_\nu\varphi - \frac{1}{2}m^2\varphi^2 \right) + \sqrt{-g} \left(-\frac{1}{2} \delta g^{\mu\nu} \nabla_\mu\varphi \nabla_\nu\varphi \right) \right] \\ &= \int d^Dx \sqrt{-g} \left[\frac{1}{2}g^{\alpha\beta} \delta g_{\alpha\beta} \left(-\frac{1}{2}g^{\mu\nu} \nabla_\mu\varphi \nabla_\nu\varphi - \frac{1}{2}m^2\varphi^2 \right) + \frac{1}{2}g^{\mu\alpha} g^{\nu\beta} \delta g_{\alpha\beta} \nabla_\mu\varphi \nabla_\nu\varphi \right] \\ &= \frac{1}{2} \int d^Dx \sqrt{-g} \delta g_{\alpha\beta} \left[\nabla^\alpha\varphi \nabla^\beta\varphi - \frac{1}{2}g^{\alpha\beta} (\nabla_\rho\varphi \nabla^\rho\varphi + m^2\varphi^2) \right] \\ &= \frac{1}{2} \int d^Dx \sqrt{-g} \delta g_{\mu\nu} \left[\nabla^\mu\varphi \nabla^\nu\varphi - \frac{1}{2}g^{\mu\nu} (\nabla_\rho\varphi \nabla^\rho\varphi + m^2\varphi^2) \right]. \end{aligned}$$

By definition, we have:

$$T^{\mu\nu} = -\nabla^\mu\varphi \nabla^\nu\varphi + \frac{1}{2}g^{\mu\nu} ((\nabla\varphi)^2 + m^2\varphi^2)$$

The symmetric of this quantity is obviously. □

4. Trace. For $m = 0$, compute $T^\mu{}_\mu$ in D dimensions for the minimally coupled scalar. Determine for which D it vanishes classically (without improvement).

Solution. (25.4)

$$T^\mu{}_\mu|_{m^2=0} = -(\nabla\varphi)^2 + \frac{1}{2}\delta^\mu{}_\mu(\nabla\varphi)^2 = \frac{D-2}{2}(\nabla\varphi)^2$$

It is clear that $T^\mu{}_\mu$ only vanishes classically in **two dimension**. □

5. Improvement term. Add the improvement $\Delta S = \int d^D x \sqrt{-g} \xi R \varphi^2$. In flat space, the improved stress tensor shifts by a total-derivative term. Compute the trace $T^\mu{}_\mu$ for $m = 0$ and determine ξ such that the trace vanishes classically.

Solution. (25.5) We first calculate the variation of ΔS

$$\begin{aligned} \delta(\Delta S) &= \xi \int d^D x \delta(\sqrt{-g} R \varphi^2) \\ &= \xi \int d^D x [\delta(\sqrt{-g}) R \varphi^2 + \sqrt{-g} \delta R \varphi^2] \\ &= \xi \int d^D x \sqrt{-g} \left[-\frac{1}{2} g_{\mu\nu} \delta g^{\mu\nu} R \varphi^2 + \delta R \varphi^2 \right] \\ &= \xi \int d^D x \sqrt{-g} \left[-\frac{1}{2} g_{\mu\nu} R \varphi^2 \delta g^{\mu\nu} + \varphi^2 (R_{\mu\nu} \delta g^{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) \delta g^{\mu\nu}) \right] \\ &= \xi \int d^D x \sqrt{-g} \left[\left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) \varphi^2 \delta g^{\mu\nu} + \varphi^2 (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) \delta g^{\mu\nu} \right] \\ &= \xi \int d^D x \sqrt{-g} [G_{\mu\nu} \varphi^2 \delta g^{\mu\nu} + \varphi^2 g_{\mu\nu} \square \delta g^{\mu\nu} - \varphi^2 \nabla_\mu \nabla_\nu \delta g^{\mu\nu}] \\ &\simeq \xi \int d^D x \sqrt{-g} [G_{\mu\nu} \varphi^2 \delta g^{\mu\nu} + g_{\mu\nu} (\square \varphi^2) \delta g^{\mu\nu} - (\nabla_\mu \nabla_\nu \varphi^2) \delta g^{\mu\nu}] \\ &= \xi \int d^D x \sqrt{-g} [G_{\mu\nu} \varphi^2 + g_{\mu\nu} \square \varphi^2 - \nabla_\mu \nabla_\nu \varphi^2] \delta g^{\mu\nu}. \end{aligned}$$

Here, we used the following Palatini equation:

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}, \quad \delta R = R_{\mu\nu}\delta g^{\mu\nu} + (g_{\mu\nu}\square - \nabla_\mu \nabla_\nu)\delta g^{\mu\nu}$$

which implies:

$$T_\Delta^{\mu\nu} := -\frac{2}{\sqrt{-g}} \frac{\delta(\Delta S)}{\delta g_{\mu\nu}} = 2\xi [G^{\mu\nu} \varphi^2 + g^{\mu\nu} \square \varphi^2 - \nabla^\mu \nabla^\nu \varphi^2]$$

Next, we compute the trace:

$$\begin{aligned} T_\Delta^\mu{}_\mu &= 2\xi [g_{\mu\nu} G^{\mu\nu} \varphi^2 + g_{\mu\nu} g^{\mu\nu} \square \varphi^2 - g_{\mu\nu} \nabla^\mu \nabla^\nu \varphi^2] \\ &= 2\xi \left[\left(R - \frac{1}{2} g_{\mu\nu} g^{\mu\nu} R \right) \varphi^2 + D \square \varphi^2 - \nabla_\mu \nabla^\mu \varphi^2 \right] \\ &= 2\xi \left[\left(1 - \frac{D}{2} \right) R \varphi^2 + (D-1) \square \varphi^2 \right] \\ &= \xi(2-D) R \varphi^2 + 2\xi(D-1) \square \varphi^2. \end{aligned}$$

Now, come back to the flat spacetime $R = 0$, we have:

$$T^\mu{}_\mu + T_\Delta^\mu{}_\mu = \frac{D-2}{2}(\partial\varphi)^2 + 2\xi(D-1)\partial^2\varphi^2 = \left(4\xi(D-1) + \frac{D-2}{2} \right) (\partial\varphi)^2 + \cancel{4\xi(D-1)\varphi\square\varphi} \xrightarrow{\text{OS: 0}}$$

so we need to choose:

$$\xi = -\frac{D-2}{8(D-1)}$$

□

6. Stress tensor conservation as Ward identity. With a source $J_{\mu\nu}(x)$ coupled as $\Delta S = \frac{1}{2} \int d^D x J_{\mu\nu} T^{\mu\nu}$, derive the momentum-space Ward identity for the 2-point function $\langle TT \rangle$ that encodes $\partial_\mu T^{\mu\nu} = 0$.

7. Weyl variation and tracelessness. For an infinitesimal Weyl rescaling $\delta g_{\mu\nu} = 2\sigma g_{\mu\nu}$, show

$$\delta S = - \int d^D x \sqrt{-g} \sigma T^\mu{}_\mu.$$

Apply this to the improved massless scalar at the conformal value of ξ and conclude $\delta S = 0$ at the classical level.

Solution. (25.7) Using the definition of energy-momentum tensor, we have:

$$T^{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S}{\delta g_{\mu\nu}} \Rightarrow \delta S = \int d^D x \frac{\delta S}{\delta g_{\mu\nu}} \delta g_{\mu\nu} = -\frac{1}{2} \int d^D x \sqrt{-g} T^{\mu\nu} \delta g_{\mu\nu}$$

send $\delta g_{\mu\nu} = 2\sigma g_{\mu\nu}$

$$\delta S = - \int d^D x \sqrt{-g} \sigma g_{\mu\nu} T^{\mu\nu} = - \int d^D x \sqrt{-g} \sigma T^\mu{}_\mu$$

Applying the value of ξ obtained in Problem 5, the classical trace $T^\mu{}_\mu$ vanishes. Consequently, the Weyl variation δS_{Weyl} also vanishes at the classical level. In other words, there is no Weyl anomaly at the classical level. □

Solution Manual of Problem Set 26

Bufan Zheng
bufan@ecc.u-tokyo.ac.jp

1. Scalar in curved space: diff invariance. For

$$S = \int d^D x \sqrt{-g} \left(\frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi - \frac{m^2}{2} \varphi^2 \right),$$

verify that under an infinitesimal diffeomorphism generated by ξ^μ , the variation is a total derivative. Compute $\delta\varphi = \xi^\mu \partial_\mu \varphi$ and $\delta g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu$.

Solution. (26.1) First, let us compute the variation of the Lagrangian:

$$\begin{aligned} \delta\mathcal{L} &= \delta \left(\frac{1}{2} g^{\mu\nu} \nabla_\mu \varphi \nabla_\nu \varphi \right) - \delta \left(\frac{m^2}{2} \varphi^2 \right) \\ &= \frac{1}{2} (\delta g^{\mu\nu}) \nabla_\mu \varphi \nabla_\nu \varphi + g^{\mu\nu} \delta(\nabla_\mu \varphi) \nabla_\nu \varphi - m^2 \varphi \delta\varphi \\ &= \frac{1}{2} [-(\nabla^\mu \xi^\nu + \nabla^\nu \xi^\mu)] \nabla_\mu \varphi \nabla_\nu \varphi + g^{\mu\nu} \nabla_\mu (\delta\varphi) \nabla_\nu \varphi - m^2 \varphi (\xi^\rho \nabla_\rho \varphi) \\ &= -(\nabla^\mu \xi^\nu) \nabla_\mu \varphi \nabla_\nu \varphi + g^{\mu\nu} \nabla_\mu (\xi^\rho \nabla_\rho \varphi) \nabla_\nu \varphi + \xi^\rho \nabla_\rho \left(-\frac{m^2}{2} \varphi^2 \right) \\ &= -(\nabla^\mu \xi^\nu) \nabla_\mu \varphi \nabla_\nu \varphi + g^{\mu\nu} [(\nabla_\mu \xi^\rho) \nabla_\rho \varphi + \xi^\rho \nabla_\mu \nabla_\rho \varphi] \nabla_\nu \varphi + \xi^\rho \nabla_\rho \left(-\frac{m^2}{2} \varphi^2 \right) \\ &= -(\nabla^\mu \xi^\nu) \nabla_\mu \varphi \nabla_\nu \varphi + (\nabla^\mu \xi^\rho) \nabla_\rho \varphi \nabla_\mu \varphi + \xi^\rho g^{\mu\nu} \nabla_\mu \nabla_\rho \varphi \nabla_\nu \varphi + \xi^\rho \nabla_\rho \left(-\frac{m^2}{2} \varphi^2 \right) \\ &= \xi^\rho g^{\mu\nu} \nabla_\mu \nabla_\rho \varphi \nabla_\nu \varphi + \xi^\rho \nabla_\rho \left(-\frac{m^2}{2} \varphi^2 \right) \\ &= \xi^\rho \nabla_\rho \left(\frac{1}{2} g^{\mu\nu} \nabla_\mu \varphi \nabla_\nu \varphi \right) + \xi^\rho \nabla_\rho \left(-\frac{m^2}{2} \varphi^2 \right) \\ &= \xi^\rho \nabla_\rho \mathcal{L}. \end{aligned}$$

Next, the variation of action is:

$$\begin{aligned} \delta S &= \delta \int d^D x \sqrt{-g} \mathcal{L} = \int d^D x \delta(\sqrt{-g} \mathcal{L}) \\ &= \int d^D x [(\delta\sqrt{-g}) \mathcal{L} + \sqrt{-g} \delta\mathcal{L}] \\ &= \int d^D x \left[\frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} \mathcal{L} + \sqrt{-g} \delta\mathcal{L} \right] \\ &= \int d^D x \left[\frac{1}{2} \sqrt{-g} g^{\mu\nu} (\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu) \mathcal{L} + \sqrt{-g} \delta\mathcal{L} \right] \\ &= \int d^D x [\sqrt{-g} (\nabla_\mu \xi^\mu) \mathcal{L} + \sqrt{-g} \delta\mathcal{L}]. \end{aligned}$$

Combining this two results, we obtain:

$$\begin{aligned}
\delta S &= \int d^D x \sqrt{-g} [(\nabla_\mu \xi^\mu) \mathcal{L} + \delta \mathcal{L}] \\
&= \int d^D x \sqrt{-g} [(\nabla_\mu \xi^\mu) \mathcal{L} + \xi^\mu \nabla_\mu \mathcal{L}] \\
&= \int d^D x \sqrt{-g} \nabla_\mu (\xi^\mu \mathcal{L}) \\
&= \int d^D x \partial_\mu (\sqrt{-g} \xi^\mu \mathcal{L}) .
\end{aligned}$$

In the last step, we used $\sqrt{-g} \nabla_\mu V^\mu = \partial_\mu (\sqrt{-g} V^\mu)$. Then we complete the proof. $\square$

2. Background gauge field as spurion. For a complex scalar with $D_\mu = \partial_\mu + iA_\mu$, show that if $\phi \rightarrow e^{i\alpha} \phi$ and $A_\mu \rightarrow A_\mu - \partial_\mu \alpha$, then $|D_\mu \phi|^2$ is invariant.

Solution. (26.2)

$$\begin{aligned}
D'_\mu \phi' &= (\partial_\mu + iA'_\mu)(e^{i\alpha} \phi) = \partial_\mu (e^{i\alpha} \phi) + i(A_\mu - \partial_\mu \alpha) e^{i\alpha} \phi \\
&= e^{i\alpha} (i\partial_\mu \alpha) \phi + e^{i\alpha} \partial_\mu \phi + i(A_\mu - \partial_\mu \alpha) e^{i\alpha} \phi = e^{i\alpha} (\partial_\mu + iA_\mu) \phi = e^{i\alpha} D_\mu \phi.
\end{aligned}$$

which implies $|D_\mu \phi|^2$ is invariant. $\square$

3. Ward identity with background gauge field. Consider $Z[A] = \int D\phi D\phi^* \exp(iS[\phi; A])$. Derive the Ward identity

$$\nabla_\mu \langle j^\mu(x) \rangle_A = 0$$

by varying $\alpha(x)$ in the background gauge symmetry.

Solution. (26.3) We assume the measurement $D\phi$ and $D\phi^*$ is invariant under $\phi \rightarrow e^{i\alpha} \phi$, i.e. this theory is anomaly free. The generate function $Z[A]$ should be invariant under the gauge transformation $\delta A_\mu(x) = \nabla_\mu \alpha(x)$:

$$\begin{aligned}
0 &= \frac{\delta Z}{\delta A_\mu(x)} = \int D\phi D\phi^* \frac{\delta}{\delta A_\mu(x)} e^{iS[\phi; A]} \\
&= \int D\phi D\phi^* i \underbrace{\left(\frac{\delta S}{\delta A_\mu(x)} \right)}_{j^\mu} e^{iS[\phi; A]} \\
&= \int d^4x (i \langle j^\mu(x) \rangle_A Z[A]) \delta A_\mu(x) \\
&= iZ[A] \int d^4x \langle j^\mu(x) \rangle_A \nabla_\mu \alpha(x) \\
&\Rightarrow \int \nabla_\mu \langle j^\mu(x) \rangle_A \alpha(x) = 0
\end{aligned}$$

Since this equation holds for any $\alpha(x)$, we obtain:

$$\nabla_\mu \langle j^\mu(x) \rangle_A = 0$$

$\square$

4. Spurionic mass term. For a Dirac fermion in curved space,

$$S = \int d^D x \sqrt{-g} \bar{\psi} (i \gamma^\mu \nabla_\mu - m) \psi,$$

treat $m(x)$ as a spurion field. Under a Weyl scaling $g_{\mu\nu} \rightarrow e^{2\sigma} g_{\mu\nu}$, choose a scaling of $m(x)$ so that the mass term scales homogeneously. What is the Weyl weight of m ?

5. Holonomy with background $U(1)$ on T^2 . On T^2 with cycles a, b , let A be flat but with holonomies $U_a = \exp(i \oint_a A)$ and $U_b = \exp(i \oint_b A)$. Show that gauge-invariant data are (U_a, U_b) modulo large gauge transformations that shift the integrals by $2\pi\mathbb{Z}$.

6. Flux sector and background metric. Let $\Sigma = S^2$ with radius R and background metric. For a $U(1)$ background with $\int_{S^2} F = 2\pi k$, take F proportional to the volume form. Compute $\int d^2 x \sqrt{g} F^{\mu\nu} F_{\mu\nu}$ in terms of k, R .

7. Spurionic general covariance and stress tensor. Couple a QFT to $g_{\mu\nu}(x)$. Show that diffeomorphism invariance implies the Ward identity

$$\nabla_\mu \langle T^{\mu\nu} \rangle = 0$$

in the absence of gravitational anomalies, by varying the generating functional under $\delta g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu$. If the theory is also coupled to a background $U(1)$ gauge field A_μ , show the Ward identity generalizes to

$$\nabla_\mu \langle T^\mu{}_\nu \rangle = \langle j^\mu \rangle F_{\mu\nu}.$$

Solution Manual of Problem Set 27

Bufan Zheng
bufan@ecc.u-tokyo.ac.jp

1. Weyl scaling of the metric. Under $g_{\mu\nu} \rightarrow e^{2\sigma(x)} g_{\mu\nu}$, compute how $\sqrt{-g}$ and $g^{\mu\nu}$ transform to first order in σ .

Solution. (27.1)

$$g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu_\nu \Rightarrow g^{\mu\nu} \rightarrow e^{-2\sigma} g^{\mu\nu} = (1 - 2\sigma) g^{\mu\nu} + O(\sigma^2) \Rightarrow \boxed{\delta g^{\mu\nu} = -2\sigma g^{\mu\nu}}$$

$$\det g_{\mu\nu} := g \rightarrow e^{2D\sigma} g \Rightarrow \sqrt{-g} \rightarrow e^{D\sigma} \sqrt{-g} = (1 + D\sigma) \sqrt{-g} + O(\sigma^2) \Rightarrow \boxed{\delta \sqrt{-g} = D\sigma \sqrt{-g}}.$$

Where D is the dimension of the spacetime. □

2. Weyl weight of a scalar kinetic term. For $S = \int d^D x \sqrt{-g} \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi$, choose a Weyl transformation $\varphi \rightarrow e^{w\sigma} \varphi$ and determine w so that the kinetic term is Weyl invariant for a global rescaling $\sigma = \text{const.}$

Solution. (27.2) From the discussion in the last problem, we know that

$$\sqrt{-g} \rightarrow e^{D\sigma} \sqrt{-g}, \quad g^{\mu\nu} \rightarrow e^{-2\sigma} g^{\mu\nu}, \quad \partial\varphi\partial\varphi \rightarrow e^{2w\sigma} \partial\varphi\partial\varphi$$

which implies:

$$S \rightarrow e^{(D-2+2w)\sigma} S$$

To be Weyl invariant, $D - 2 + 2w$ should vanish, so **MY FINAL ANSWER IS:**

$$\boxed{w = \frac{2-D}{2}}$$

□

3. Conformal coupling constant. Include $\xi R \varphi^2$. For $m = 0$, determine ξ such that the full action is Weyl invariant up to total derivatives in flat space, and hence the improved stress tensor is traceless classically.

Solution. (27.3) Using the result in the last problem, we choose $w = 1 - \frac{D}{2}$. However, here we don't assume σ is a constant, then we find the variation of kinematic term:

$$\begin{aligned} \tilde{\mathcal{L}}_{\text{kin}} &= \sqrt{-g} \frac{1}{2} \tilde{g}^{\mu\nu} \partial_\mu \tilde{\varphi} \partial_\nu \tilde{\varphi} \\ &= \sqrt{-g} e^{D\sigma} \frac{1}{2} e^{-2\sigma} g^{\mu\nu} \partial_\mu (e^{w\sigma} \varphi) \partial_\nu (e^{w\sigma} \varphi) \\ &= \sqrt{-g} \frac{1}{2} e^{(D-2)\sigma} g^{\mu\nu} (e^{w\sigma} (\partial_\mu \varphi + w \varphi \partial_\mu \sigma)) (e^{w\sigma} (\partial_\nu \varphi + w \varphi \partial_\nu \sigma)) \\ &= \sqrt{-g} \frac{1}{2} e^{(D-2+2w)\sigma} g^{\mu\nu} (\partial_\mu \varphi + w \varphi \partial_\mu \sigma) (\partial_\nu \varphi + w \varphi \partial_\nu \sigma) \\ &= \sqrt{-g} \frac{1}{2} e^{(D-2+2w)\sigma} \left((\partial\varphi)^2 + 2w \varphi \partial^\mu \sigma \partial_\mu \varphi + w^2 \varphi^2 (\partial\sigma)^2 \right) \\ &\xrightarrow{w=\frac{2-D}{2}} \sqrt{-g} \frac{1}{2} \left((\partial\varphi)^2 + (2-D) \varphi \partial^\mu \sigma \partial_\mu \varphi + \frac{(D-2)^2}{4} \varphi^2 (\partial\sigma)^2 \right) \end{aligned}$$

To continue, we need the following weyl transformation formula of Ricci scalar:

$$\begin{aligned}
\tilde{\Gamma}^\rho_{\mu\nu} &= \frac{1}{2}\tilde{g}^{\rho\lambda}\left(\partial_\mu\tilde{g}_{\lambda\nu} + \partial_\nu\tilde{g}_{\lambda\mu} - \partial_\lambda\tilde{g}_{\mu\nu}\right) \\
&= \frac{1}{2}e^{-2\sigma}g^{\rho\lambda}\left(\partial_\mu(e^{2\sigma}g_{\lambda\nu}) + \partial_\nu(e^{2\sigma}g_{\lambda\mu}) - \partial_\lambda(e^{2\sigma}g_{\mu\nu})\right) \\
&= \frac{1}{2}g^{\rho\lambda}\left(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}\right) + g^{\rho\lambda}\left(g_{\lambda\nu}\partial_\mu\sigma + g_{\lambda\mu}\partial_\nu\sigma - g_{\mu\nu}\partial_\lambda\sigma\right) \\
&= \Gamma^\rho_{\mu\nu} + \delta^\rho_\mu\partial_\nu\sigma + \delta^\rho_\nu\partial_\mu\sigma - g_{\mu\nu}\partial^\rho\sigma \equiv \Gamma^\rho_{\mu\nu} + C^\rho_{\mu\nu} \\
\tilde{R} &= \tilde{g}^{\mu\nu}\left(\partial_\rho\tilde{\Gamma}^\rho_{\mu\nu} - \partial_\nu\tilde{\Gamma}^\rho_{\mu\rho} + \tilde{\Gamma}^\rho_{\rho\lambda}\tilde{\Gamma}^\lambda_{\mu\nu} - \tilde{\Gamma}^\rho_{\nu\lambda}\tilde{\Gamma}^\lambda_{\mu\rho}\right) \\
&= e^{-2\sigma}g^{\mu\nu}\left(\partial_\rho\tilde{\Gamma}^\rho_{\mu\nu} - \partial_\nu\tilde{\Gamma}^\rho_{\mu\rho} + \tilde{\Gamma}^\rho_{\rho\lambda}\tilde{\Gamma}^\lambda_{\mu\nu} - \tilde{\Gamma}^\rho_{\nu\lambda}\tilde{\Gamma}^\lambda_{\mu\rho}\right) \\
&= e^{-2\sigma}g^{\mu\nu}\left(\partial_\rho(\Gamma^\rho_{\mu\nu} + C^\rho_{\mu\nu}) - \partial_\nu(\Gamma^\rho_{\mu\rho} + C^\rho_{\mu\rho})\right. \\
&\quad \left.+ (\Gamma^\rho_{\rho\lambda} + C^\rho_{\rho\lambda})(\Gamma^\lambda_{\mu\nu} + C^\lambda_{\mu\nu}) - (\Gamma^\rho_{\nu\lambda} + C^\rho_{\nu\lambda})(\Gamma^\lambda_{\mu\rho} + C^\lambda_{\mu\rho})\right) \\
&= e^{-2\sigma}\left(R + g^{\mu\nu}(\nabla_\rho C^\rho_{\mu\nu} - \nabla_\nu C^\rho_{\mu\rho}) + g^{\mu\nu}(C^\rho_{\rho\lambda}C^\lambda_{\mu\nu} - C^\rho_{\nu\lambda}C^\lambda_{\mu\rho})\right) \\
&= e^{-2\sigma}\left(R + \nabla_\rho(g^{\mu\nu}C^\rho_{\mu\nu}) - \nabla_\nu(g^{\mu\nu}C^\rho_{\mu\rho}) + g^{\mu\nu}(C^\rho_{\rho\lambda}C^\lambda_{\mu\nu} - C^\rho_{\nu\lambda}C^\lambda_{\mu\rho})\right) \\
&= e^{-2\sigma}\left(R + \nabla_\rho(g^{\mu\nu}(\delta^\rho_\mu\partial_\nu\sigma + \delta^\rho_\nu\partial_\mu\sigma - g_{\mu\nu}\partial^\rho\sigma)) - \nabla_\nu(g^{\mu\nu}(D\partial_\mu\sigma))\right. \\
&\quad \left.+ g^{\mu\nu}(D\partial_\lambda\sigma)C^\lambda_{\mu\nu} - g^{\mu\nu}C^\rho_{\nu\lambda}C^\lambda_{\mu\rho}\right) \\
&= e^{-2\sigma}\left(R + \nabla_\rho((2-D)\partial^\rho\sigma) - D\nabla^2\sigma + D\partial_\lambda\sigma(2-D)\partial^\lambda\sigma - (2-D)(\partial\sigma)^2\right) \\
&= e^{-2\sigma}\left(R - 2(D-1)\nabla^2\sigma - (D-1)(D-2)(\nabla\sigma)^2\right)
\end{aligned}$$

Therefore, we have:

$$\begin{aligned}
\tilde{\mathcal{L}}_\xi &= \sqrt{-\tilde{g}}\xi\tilde{R}\tilde{\varphi}^2 \\
&= \sqrt{-g}e^{D\sigma}\xi\left(e^{-2\sigma}(R - 2(D-1)\nabla^2\sigma - (D-1)(D-2)(\nabla\sigma)^2)\right)(e^{2w\sigma}\varphi^2) \\
&= \sqrt{-g}\xi e^{(D-2+2w)\sigma}\left(R - 2(D-1)\nabla^2\sigma - (D-1)(D-2)(\nabla\sigma)^2\right)\varphi^2 \\
&\xrightarrow{w=\frac{2-D}{2}} \sqrt{-g}\xi\left(R - 2(D-1)\nabla^2\sigma - (D-1)(D-2)(\nabla\sigma)^2\right)\varphi^2
\end{aligned}$$

Using

$$\sqrt{-g}\varphi\partial^\mu\sigma\partial_\mu\varphi = \frac{1}{2}\sqrt{-g}\nabla_\mu(\varphi^2\nabla^\mu\sigma) - \frac{1}{2}\sqrt{-g}\varphi^2\nabla^2\sigma,$$

we have:

$$\begin{aligned}
\tilde{\mathcal{L}}_{\text{total}} - \tilde{\mathcal{L}} &= \sqrt{-g}\varphi^2\left[\nabla^2\sigma\left(-2\xi(D-1) - \frac{2-D}{4}\right) + (\nabla\sigma)^2\left(-(D-1)(D-2)\xi + \frac{(D-2)^2}{8}\right)\right] \\
&\quad + \frac{2-D}{4}\sqrt{-g}\nabla_\mu(\varphi^2\nabla^\mu\sigma)
\end{aligned}$$

To leave only a total derivative:

$$\begin{cases} -\frac{2-D}{4} - 2\xi(D-1) = 0 \\ -(D-1)(D-2)\xi + \frac{(D-2)^2}{8} = 0 \end{cases}$$

which implies:

$$\boxed{\xi = \frac{D-2}{8(D-1)}}$$

□

4. Conformal Killing vectors. In flat space, $\delta g_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Impose $\delta g_{\mu\nu} = 2\sigma \eta_{\mu\nu}$. Solve for $\sigma = \frac{1}{D} \partial \cdot \xi$. Verify that dilations $\xi^\mu = \lambda x^\mu$ satisfy this and compute σ .

Solution. (27.4) In flat spacetime, the conformal Killing equation is:

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = 2\sigma \eta_{\mu\nu}$$

Contract both sides with $\eta^{\mu\nu}$

$$\eta^{\mu\nu}(\partial_\mu \xi_\nu + \partial_\nu \xi_\mu) = 2\sigma \eta^{\mu\nu} \eta_{\mu\nu} \Rightarrow 2\partial_\mu \xi^\mu = 2\sigma D \Rightarrow \sigma = \frac{1}{D} \partial \cdot \xi$$

For dilation, we have:

$$\partial_\mu \xi^\mu = \partial_\mu (\lambda x^\mu) = \lambda \delta^\mu_\mu = \lambda D \Rightarrow \sigma = \frac{1}{D} (\partial \cdot \xi) = \lambda$$

Indeed, it satisfies the conformal Killing equation:

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = \lambda(\eta_{\mu\nu} + \eta_{\nu\mu}) = 2\lambda \eta_{\mu\nu} = 2\sigma \eta_{\mu\nu}$$

□

5. Conformal group dimension in D . Count generators of the conformal group in $D \geq 3$: translations P , Lorentz M , dilations D , special conformal K . Give the total number as a function of D .

Solution. (27.5)

- Translations P_μ : D generators.
- Lorentz transformations $M_{\mu\nu} = -M_{\nu\mu}$: $\frac{D(D-1)}{2}$ generators.
- Dilations $\mathfrak{D}$: 1 generator.
- Special conformal transformations K_μ : D generators.

There the total number is:

$$\# = D + \frac{D(D-1)}{2} + 1 + D = \frac{(D+1)(D+2)}{2}, \quad D \geq 3$$

□

6. Weyl as spurionic symmetry. Treat couplings g_i as spacetime-dependent spurions with Weyl weights w_i . For an operator $\mathcal{O}_i$ of classical scaling dimension Δ_i , find w_i so that $\int d^D x \sqrt{-g} g_i \mathcal{O}_i$ is Weyl invariant in flat space.

7. Conformal as dynamical subgroup of $\text{Diff} \times \text{Weyl}$. In flat space, consider a diffeomorphism $x \rightarrow x + \xi(x)$ combined with a Weyl rescaling $\sigma(x)$. Demand the combined transformation preserves $\eta_{\mu\nu}$. Derive $\sigma = \frac{1}{D} \partial \cdot \xi$ and conclude the condition on ξ is the conformal Killing equation.

Solution. (27.7) An infinitesimal diffeomorphism changes the metric by the Lie derivative, namely,

$$\delta_\xi g_{\mu\nu} = \mathcal{L}_\xi g_{\mu\nu} = \xi^\rho \partial_\rho g_{\mu\nu} + (\partial_\mu \xi^\rho) g_{\rho\nu} + (\partial_\nu \xi^\rho) g_{\mu\rho}$$

in flat spacetime, we have:

$$\delta_\xi \eta_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$$

From Problem 1, we already know that the leading-order contribution to the metric under a Weyl transformation is

$$\delta_\sigma \eta_{\mu\nu} = -2\sigma \eta_{\mu\nu}$$

Combining these two facts, and requiring the two effects to cancel, we obtain

$$\boxed{\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = 2\sigma \eta_{\mu\nu}}$$

Which is exactly the conformal Killing equation. From the problem 5, we know the solution is:

$$\boxed{\sigma = \frac{1}{D} \partial \cdot \xi}$$

□

Solution Manual of Problem Set 28

Bufan Zheng

bufan@g.ecc.u-tokyo.ac.jp

Conventions: Lorentzian path integral $Z[A] = \int D\Phi \exp(iS[\Phi; A])$ and $W[A] = -i \ln Z[A]$. To avoid ambiguity in factors of i , anomalies are stated below as variations of $\ln Z$.

1. 2d $U(1)$ anomaly and large gauge transformations. In two dimensions, suppose the quantum effective action has a gauge anomaly of the form

$$\delta_\alpha \ln Z[A] = \frac{k}{2\pi} \int \alpha F$$

for infinitesimal α , where $F = dA$. Show that on a closed 2-manifold, consistency with large gauge transformations and flux quantization constrains k .

2. Anomaly inflow (3d bulk cancels 2d boundary). Consider a 3d bulk with action $S_{\text{CS}} = \frac{k}{4\pi} \int A \wedge dA$. Show that under $A \rightarrow A + d\alpha$ the bulk variation is a boundary term. Explain how this can cancel the 2d anomaly in Problem 1.

3. 4d chiral anomaly: cubic charge constraint. Consider a set of left-handed Weyl fermions of $U(1)$ charges q_i . Argue that the $U(1)^3$ gauge anomaly coefficient is proportional to $\sum_i q_i^3$. State the condition for gauge anomaly cancellation.

Solution. (28.4) The general form of $D = 4$ anomaly polynomial of abelian gauge theory is:

$$I_6 = \frac{k}{(2\pi)^3} F \wedge F \wedge F + \frac{\ell}{2\pi} F \wedge p_1(R)$$

The first term is related to the pure gauge anomaly. This term is proportional to $\mathcal{F}^3$, which is just $\sum_i q_i^3$ in the abelian gauge theory. So we know that the condition for gauge anomaly cancellation is:

$$\boxed{\sum_i q_i^3 = 0}$$

□

4. Mixed gauge-gravitational anomaly (statement level). For $U(1)$ in 4d with chiral fermions, there is also a possible mixed gauge-gravitational anomaly proportional to $\sum_i q_i$. State the corresponding cancellation condition and give a short interpretation.

Solution. (28.5) The term related to mixed gauge-gravitational anomaly is the second term in I_6 , which is proportional to

$$I_6 \supset \frac{\ell}{2\pi} F \wedge p_1(R) \propto \sum_i q_i$$

Therefore, the corresponding cancellation condition is:

$$\boxed{\sum_i q_i = 0}$$

□

5. 't Hooft anomaly matching (conceptual). Explain the meaning of an 't Hooft anomaly for a global symmetry. Why must the anomaly match between UV and IR descriptions, even across strongly coupled RG flow?

6. Discrete remnant after Higgsing with an anomaly-free parent. Suppose a $U(1)$ gauge theory is Higgsed by a charge- N scalar, leaving a $\mathbb{Z}_N$ remnant. Discuss when a $\mathbb{Z}_N$ anomaly can appear, and how it is constrained by embedding into an anomaly-free $U(1)$.

7. Background-field viewpoint. Recast anomaly cancellation conditions in terms of background gauge invariance of $Z[A]$. Explain why the anomaly coefficient is physical (scheme independent) for gauge symmetries.

Solution Manual of Problem Set 29

Bufan Zheng

bufan@g.ecc.u-tokyo.ac.jp

1. Coarse graining in 0+1d. Consider the Euclidean path integral for a single degree of freedom with a UV cutoff Λ . Split the field into slow and fast modes in a small momentum shell. Integrate out the fast modes to obtain an effective action for the slow modes.

2. Induced operators. Starting from a quartic interaction, show that integrating out a shell generates additional local operators (for example a mass term and higher-derivative terms). Identify which are relevant, marginal, and irrelevant by engineering dimension.

3. Rescaling step. After integrating out a momentum shell, rescale coordinates and fields to restore the cutoff to Λ . Derive the classical scaling of couplings under this step.

4. Fixed points and linearization. Define what it means for an action (or set of couplings) to be at a fixed point of the RG. Linearize the flow around a fixed point and define critical exponents in terms of eigenvalues of the linearized RG map.

Solution. (29.4) Let the Wilsonian action be parameterized by couplings $\{g_i\}$ at cutoff Λ ,

$$S_\Lambda[\phi] = S[\phi; \{g_i(\Lambda)\}], \quad t \equiv \ln \frac{\Lambda}{\Lambda_0}.$$

The RG flow is

$$\frac{dg_i}{dt} = \beta_i(\{g\}).$$

A fixed point $\{g_i^*\}$ is defined by

$$\boxed{\beta_i(\{g^*\}) = 0 \quad \forall i},$$

so that under an RG step the action keeps the same functional form (up to the trivial rescalings built into the RG transformation). Write $g_i(t) = g_i^* + \delta g_i(t)$. Expanding the beta functions,

$$\beta_i(\{g\}) = \beta_i(\{g^*\}) + \sum_j \left. \frac{\partial \beta_i}{\partial g_j} \right|_{g^*} \delta g_j + O(\delta g^2) = \sum_j M_{ij} \delta g_j + O(\delta g^2),$$

Keeping only the linear terms gives

$$\frac{d(\delta g_i)}{dt} = \sum_j M_{ij} \delta g_j.$$

Diagonalize M by choosing eigenvectors $v^{(a)}$ and eigenvalues λ_a , then directions in RG flow can be classified by the sign of λ_a :

- $y_a > 0$: relevant
- $y_a < 0$: marginal (need higher order)
- $y_a = 0$: irrelevant

Let u be a single relevant scaling variable with eigenvalue $y > 0$ so that $u(t) = u(0)e^{yt}$. Under the RG rescaling by a factor $b = e^{-t}$, correlation lengths satisfy $\xi(u) = b\xi(u(t))$. Choose t such that $u(t) \sim O(1)$, i.e. $e^{yt} \sim 1/u(0)$, so $b = e^{-t} \sim u(0)^{1/y}$. Then we obtain the definition of critical exponents ν :

$$\xi(u) \sim u(0)^{-1/y} \implies \boxed{\nu = \frac{1}{y}}$$

□

5. Universality. Explain why irrelevant operators do not affect long-distance physics at a critical point. Give a simple example illustrating universality classes.

6. Nonperturbative RG as a functional flow. Write schematically what it means to have an RG flow equation for the full Wilsonian effective action $S_\Lambda[\phi]$ rather than for finitely many couplings. Explain why such an equation can encode nonperturbative information.

7. Infinitesimal Wilsonian RG has one-loop form. Derive the infinitesimal RG step $\Lambda \rightarrow \Lambda - \delta\Lambda$ by integrating out an infinitesimal momentum shell of (approximately) Gaussian modes. Show that the exact flow equation for S_Λ (or equivalently $\mathcal{L}_{\text{eff}}$) has a one-loop functional form (a single shell contraction or single functional trace/logdet structure). Explain briefly why expanding the flowing action in couplings can nevertheless reproduce multi-loop beta functions when iterated.

Solution Manual of Problem Set 30

Bufan Zheng
bufan@g.ecc.u-tokyo.ac.jp

1. $D = 4$: Review the computation of the QED β -function. Show that for n_f Dirac fermions,

$$\beta(e) \equiv \mu \frac{de}{d\mu} = \frac{e^3}{16\pi^2} \frac{4}{3} n_f.$$

2. $D = 4$: Show that the β -function for e^2 is

$$\beta(e^2) \equiv \mu \frac{de^2}{d\mu} = \frac{e^4}{16\pi^2} \frac{8}{3} n_f.$$

Solution. (30.2)

$$\beta(e^2) = \mu \frac{de^2}{d\mu} = \mu 2e \frac{de}{d\mu} = \frac{e^4}{16\pi^2} \frac{8}{3} n_f$$

□

3. $D = 4$: Define $\alpha \equiv e^2/(4\pi)$. Show that

$$\beta(\alpha) \equiv \mu \frac{d\alpha}{d\mu} = \frac{2}{3\pi} n_f \alpha^2.$$

Solution. (30.3)

$$\beta(\alpha) = \mu \frac{d\alpha}{d\mu} = \frac{1}{4\pi} \cdot \mu \frac{de^2}{d\mu} = \frac{\beta(e^2)}{4\pi} = \frac{2}{3\pi} n_f \frac{e^4}{16\pi^2} = \frac{2}{3\pi} n_f \alpha^2$$

□

4. $D = 4$: In the modern normalization of the gauge field, the Lagrangian contains

$$\mathcal{L} \supset -\frac{1}{4e^2} F^{\mu\nu} F_{\mu\nu}.$$

Show that

$$\mu \frac{d}{d\mu} \left(\frac{1}{e^2} \right) = -\frac{1}{16\pi^2} \frac{8}{3} n_f.$$

Solution. (30.4)

$$\mu \frac{d}{d\mu} \left(\frac{1}{e^2} \right) = -\frac{1}{e^4} \cdot \mu \frac{de^2}{d\mu} = -\frac{\beta(e^2)}{e^4} = -\frac{1}{16\pi^2} \frac{8}{3} n_f$$

□

5. $D = 4 - \epsilon$: Use dimensional analysis to show that classically

$$\beta(e) = \frac{D-4}{2} e.$$

Combine this with the one-loop term to write the full β -function to leading order in e and ϵ .

Solution. (30.5) The derivation of $\beta_{\text{cl.}}(e)$ has been done in problem 22.7, so here we will skip the derivation and directly quote the result:

$$\beta_{\text{cl.}}(e) = \frac{D-4}{2}e$$

Thus for $D = 4 - \epsilon$,

$$\beta_{\text{cl.}}(e) = -\frac{\epsilon}{2}e$$

For QED with n_f Dirac fermions, the one-loop contribution in four dimensions is

$$\beta_{1\text{-loop}}(e) = \frac{n_f}{12\pi^2}e^3.$$

Keeping the leading terms in both ϵ and e in $D = 4 - \epsilon$ gives

$$\boxed{\beta(e) = -\frac{\epsilon}{2}e + \frac{n_f}{12\pi^2}e^3 + \dots}$$

□

6. $D = 4 - \epsilon$: Define $\hat{e}^2 \equiv e^2\mu^{-\epsilon}$. Show that the β -function for $\hat{e}^2$ takes the form

$$\beta(\hat{e}^2) = -\epsilon\hat{e}^2 + \frac{8}{3}\frac{n_f}{16\pi^2}\hat{e}^4.$$

Find the nontrivial fixed point $\hat{e}_*^2$ and determine whether it is IR or UV attractive.

Solution. (30.6)

$$\begin{aligned}\beta(\hat{e}^2) &= \mu \frac{d\hat{e}^2}{d\mu} = \mu \frac{d}{d\mu}(e^2\mu^{-\epsilon}) = \mu^{-\epsilon}\mu \frac{de^2}{d\mu} - \epsilon e^2\mu^{-\epsilon} \\ &= \mu^{-\epsilon}\beta(e^2) - \epsilon\hat{e}^2 = \mu^{-\epsilon}(2e\beta(e)) - \epsilon\hat{e}^2 = \mu^{-\epsilon}\left(2e\frac{n_f}{12\pi^2}e^3\mu^{-\epsilon}\right) - \epsilon\hat{e}^2 \\ &= -\epsilon\hat{e}^2 + \frac{n_f}{6\pi^2}e^4\mu^{-2\epsilon} = -\epsilon\hat{e}^2 + \frac{n_f}{6\pi^2}(e^2\mu^{-\epsilon})^2 = -\epsilon\hat{e}^2 + \frac{n_f}{6\pi^2}\hat{e}^4 \\ &= -\epsilon\hat{e}^2 + \frac{8}{3}\frac{n_f}{16\pi^2}\hat{e}^4.\end{aligned}$$

The $\beta(e)$ we obtained earlier, in the notation used here, should actually be $\beta(\hat{e})$, i.e. the beta function of the dimensionless coupling. However, the $\beta(e)$ we need to substitute above is the beta function of the dimensionful coupling. Since $[e] = \frac{\epsilon}{2}$, we multiply the previously obtained beta function by $\mu^{-\epsilon}$ to supply the missing mass dimension. Solving $\beta(\hat{e}^2) = 0$ yields

$$\hat{e}_*^2 = 0, \quad \hat{e}_*^2 = \epsilon \left[\frac{8}{3} \frac{n_f}{16\pi^2} \right]^{-1} = \boxed{\hat{e}_*^2 = \frac{6\pi^2}{n_f} \epsilon}.$$

We focus on the boxed non-trivial fixed point. Linearizing, $\beta(x) \simeq \beta'(x_*)(x - x_*)$ with

$$\beta'(x) = -\epsilon + 2\frac{8}{3}\frac{n_f}{16\pi^2}x.$$

Hence

$$\beta'(\hat{e}_*^2) = -\epsilon + 2\epsilon = +\epsilon > 0 \quad \Rightarrow \quad \hat{e}^2 = \hat{e}_*^2 \text{ is IR-attractive.}$$

So we conclude

$$\boxed{\text{For } \epsilon > 0 : \hat{e}_*^2 = \frac{6\pi^2}{n_f}\epsilon \text{ is IR-attractive.}}$$

□

7. $D = 4$: Suppose you have n_f Dirac fermions of charges q_i and n_s complex scalars of charges Q_a . Show that the one-loop coefficient becomes

$$\beta(e) = \frac{e^3}{16\pi^2} \left(\frac{4}{3} \sum_{i=1}^{n_f} q_i^2 + \frac{1}{3} \sum_{a=1}^{n_s} Q_a^2 \right).$$

Solution Manual of Problem Set 31

Bufan Zheng

bufan@ecc.u-tokyo.ac.jp

Conventions: gauge group G with generators T^a in a representation R normalized by

$$\text{tr}_R(T^a T^b) = T(R) \delta^{ab}, \quad (T^a T^a) = C(R) \mathbf{1}, \quad f^{acd} f^{bcd} = C_A \delta^{ab}.$$

Modern normalization: $\mathcal{L} \supset -\frac{1}{4g^2} F^{\mu\nu a} F_{\mu\nu}^a$.

1. One-loop structure and the sign. State the one-loop β -function for pure Yang–Mills in $D = 4$ in the form

$$\beta(g) \equiv \mu \frac{dg}{d\mu} = -\frac{g^3}{16\pi^2} \frac{11}{3} C_A.$$

Explain in one sentence why the sign is opposite to QED (screening vs antiscreening).

Solution. (31.1)

Explain in one sentence. The self-interactions of non-Abelian gauge fields make the vacuum polarization antiscreening, so the coupling decreases at high energies, whereas in QED the fluctuations of charged matter screen the charge, so the coupling increases at high energies. $\square$

2. Including matter (general formula). For n_f Dirac fermions in representations R_f and n_s complex scalars in representations R_s , show that the one-loop coefficient can be written as

$$\beta(g) = -\frac{g^3}{16\pi^2} b_0, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} \sum_{\text{Dirac } f} T(R_f) - \frac{1}{6} \sum_{\text{cpx } s} T(R_s).$$

3. Asymptotic freedom condition. Using Problem 2, give the condition on the matter content for asymptotic freedom. Apply it to $G = SU(N)$ with n_f Dirac fermions in the fundamental.

Solution. (31.3) Asymptotic freedom means:

$$b_0 = \frac{11}{3} C_A - \frac{4}{3} \sum_{\text{Dirac } f} T(R_f) - \frac{1}{6} \sum_{\text{cpx } s} T(R_s) > 0$$

For $SU(N)$ group, we have:

$$C_A = N, \quad T(R_f) = \frac{1}{2}$$

Hence

$$b_0 = \frac{11}{3} \times N - \frac{4}{3} \times n_f \times \frac{1}{2} > 0 \Rightarrow n_f < \frac{11}{2} N$$

For QCD $N = 3$ and $n_f = 6$, so we know that it is Asymptotic free. $\square$

4. Running coupling and dimensional transmutation. Solve the one-loop RG equation in $D = 4$ for $g(\mu)$ in terms of a reference scale μ_0 and $g(\mu_0)$. Define the RG-invariant scale Λ and rewrite $g(\mu)$ in terms of Λ (dimensional transmutation).

Solution. The 1-loop RG equation is:

$$\mu \frac{dg}{d\mu} = -\frac{b_0}{16\pi^2} g^3$$

which can be solved by

$$\frac{1}{g^2(\mu)} = \frac{1}{g^2(\mu_0)} + \frac{b_0}{8\pi^2} \ln \frac{\mu}{\mu_0}.$$

Define the RG-invariant scale Λ by

$$\Lambda \equiv \mu \exp\left(-\frac{8\pi^2}{b_0 g^2(\mu)}\right).$$

By construction, Λ is independent of μ . Using this definition, the running coupling can be rewritten as

$$\frac{1}{g^2(\mu)} = \frac{b_0}{8\pi^2} \ln \frac{\mu}{\Lambda} \iff \boxed{g^2(\mu) = \frac{8\pi^2}{b_0 \ln(\mu/\Lambda)}}.$$

This is dimensional transmutation: the classically dimensionless coupling g is traded, through quantum effects, for a dimensionful scale Λ . $\square$

5. Modern normalization: flow of $1/g^2$. Starting from $\beta(g) = -\frac{b_0}{16\pi^2} g^3$, show that

$$\mu \frac{d}{d\mu} \left(\frac{1}{g^2} \right) = \frac{b_0}{8\pi^2}.$$

Explain why this is the natural quantity to quote when the action is written with $-\frac{1}{4g^2} F^2$.

Solution. (31.5)

$$\mu \frac{d}{d\mu} \left(\frac{1}{g^2} \right) = \mu \left(-2g^{-3} \frac{dg}{d\mu} \right) = -2g^{-3} \left(\mu \frac{dg}{d\mu} \right) = -2g^{-3} \beta(g) = -2g^{-3} \left(-\frac{b_0}{16\pi^2} g^3 \right) = \frac{b_0}{8\pi^2}.$$

The parameter $1/g^2$ multiplies the gauge kinetic term directly, so it appears linearly as the coefficient in front of F^2 . $\square$

6. $D = 4 - \epsilon$: classical scaling vs one-loop. Use dimensional analysis to show that classically

$$\beta(g) = \frac{D-4}{2} g,$$

and combine with the one-loop term to write

$$\beta(g) = \frac{D-4}{2} g - \frac{b_0}{16\pi^2} g^3$$

to leading order in g and ϵ . Discuss when a nontrivial fixed point can occur.

Solution. (31.6) Take the Yang–Mills action in the modern normalization

$$S = -\frac{1}{4g^2} \int d^D x F_{\mu\nu}^a F^{a\mu\nu}, \quad F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c.$$

In units where $\hbar = c = 1$, the action is dimensionless, so the Lagrangian density has mass dimension

$$[\mathcal{L}] = D.$$

Also $[\partial_\mu] = 1$. From the kinetic term $(\partial A)^2 \subset F^2$ we get

$$[F_{\mu\nu}] = [\partial A] = 1 + [A].$$

Hence

$$[F_{\mu\nu}^2] = 2[F_{\mu\nu}] = 2(1 + [A]).$$

Now impose $[\mathcal{L}] = D$ for $\mathcal{L} \sim \frac{1}{g^2} F^2$:

$$\left[\frac{1}{g^2} \right] + [F^2] = D \implies -2[g] + 2(1 + [A]) = D.$$

To fix $[A]$, we back to the old normalization and use the fact that the pure gauge kinetic term is the quadratic piece $(\partial A)^2$, which must already have dimension D , giving

$$[A] = \frac{D-2}{2}.$$

Plugging this into the previous relation yields

$$-2[g] + 2\left(1 + \frac{D-2}{2}\right) = D \implies -2[g] + D = D \implies [g] = \frac{4-D}{2}.$$

Because g has mass dimension $(4-D)/2$, the dimensionless coupling is

$$\hat{g} \equiv \mu^{\frac{D-4}{2}} g.$$

At the classical level (no quantum corrections), $\hat{g}$ is scale independent, so

$$0 = \mu \frac{d\hat{g}}{d\mu} = \mu \frac{d}{d\mu} \left(\mu^{\frac{D-4}{2}} g \right) = \frac{D-4}{2} \hat{g} + \mu^{\frac{D-4}{2}} \beta(g).$$

Multiplying by $\mu^{\frac{4-D}{2}}$ gives the classical beta function for g :

$$\beta_{\text{cl}}(g) = \frac{D-4}{2} g.$$

Including the one-loop correction, we have,

$$\beta(g) = \frac{D-4}{2} g - \frac{b_0}{16\pi^2} g^3$$

In particular, for $D = 4 - \epsilon$,

$$\beta(g) = -\frac{\epsilon}{2} g - \frac{b_0}{16\pi^2} g^3 \quad (D = 4 - \epsilon).$$

Fixed points satisfy $\beta(g_*) = 0$. Besides the trivial fixed point $g_* = 0$, a nontrivial fixed point would require

$$\frac{D-4}{2} - \frac{b_0}{16\pi^2} g_*^2 = 0 \implies g_*^2 = \frac{8\pi^2}{b_0} (D-4).$$

Thus a real (physical) nontrivial fixed point exists only if

$$\frac{D-4}{b_0} > 0,$$

□

7. Banks–Zaks type regime (parametric). For $SU(N)$ with n_f fundamentals, treat n_f near the loss-of-asymptotic-freedom value so that b_0 is small and positive. Using the $D = 4 - \epsilon$ form from Problem 6 with $\epsilon > 0$ small, estimate the fixed-point coupling g_*^2 scaling with b_0 and ϵ . (You do not need the two-loop coefficient.)

Solution. (31.7) The non-trivial fixed point has been solved in the last problem. For $D = 4 - \epsilon$, we have,

$$g_*^2 = -\frac{8\pi^2}{b_0}\epsilon < 0,$$

since $\epsilon > 0$ and $b_0 > 0$. So we conclude, without any knowledge about 2-loop correction, the fixed-point is parametric. $\square$

Solution Manual of Problem Set 32

Bufan Zheng
bufan@g.ecc.u-tokyo.ac.jp

1. One-loop running in QED. Assume $\beta(\alpha) = b\alpha^2$ with $b > 0$. Solve for $\alpha(\mu)$ in terms of $\alpha(\mu_0)$ and μ/μ_0 .

Solution. (32.1)

$$\beta(\alpha) = \mu \frac{d\alpha}{d\mu} = b\alpha^2 \Rightarrow \frac{d\alpha}{\alpha^2} = b \frac{d\mu}{\mu} = b d \ln \mu \Rightarrow \frac{1}{\alpha(\mu)} = \frac{1}{\alpha(\mu_0)} - b \ln \frac{\mu}{\mu_0}$$

So, **MY FINAL ANSWER IS**

$$\alpha(\mu) = \frac{\alpha(\mu_0)}{1 - b\alpha(\mu_0) \ln(\mu/\mu_0)}.$$

□

2. Landau pole scale. Define the Landau pole Λ_{Landau} as the scale where $\alpha(\mu)$ diverges in the one-loop solution. Express Λ_{Landau} in terms of μ_0 and $\alpha(\mu_0)$.

Solution. (32.2) The Landau pole Λ_{Landau} is defined as the scale where the one-loop solution for $\alpha(\mu)$ diverges,

$$1 - b\alpha(\mu_0) \ln \frac{\Lambda_{\text{Landau}}}{\mu_0} = 0 \Rightarrow \ln \frac{\Lambda_{\text{Landau}}}{\mu_0} = \frac{1}{b\alpha(\mu_0)}.$$

Hence, **MY FINAL ANSWER IS**

$$\Lambda_{\text{Landau}} = \mu_0 \exp\left(\frac{1}{b\alpha(\mu_0)}\right).$$

□

3. Meaning of the Landau pole. Explain why a Landau pole indicates the theory is not UV complete as a fundamental theory, but can still be predictive as an effective field theory at energies well below Λ_{Landau} .

Solution. (32.3) A Landau pole means the perturbative RG flow drives the coupling to infinity at a finite energy scale, so the theory cannot be consistently extrapolated to arbitrarily high energies as a fundamental UV-complete description without new physics entering before Λ_{Landau} . Nevertheless, for $\mu \ll \Lambda_{\text{Landau}}$ the coupling remains small and perturbation theory is reliable, so the theory can be predictive as an effective field theory at energies well below Λ_{Landau} , with UV-sensitive effects suppressed by powers of $\mu/\Lambda_{\text{Landau}}$. □

4. Choosing an EFT cutoff below the pole. Let m be a light mass scale in the EFT and define

$$\Lambda_{\text{cutoff}} \equiv \sqrt{m \Lambda_{\text{Landau}}}.$$

Show that $m \ll \Lambda_{\text{cutoff}} \ll \Lambda_{\text{Landau}}$ when $\Lambda_{\text{Landau}} \gg m$. Interpret this choice.

Solution. (32.4) If $\Lambda_{\text{Landau}} \gg m$, then

$$\frac{\Lambda_{\text{cutoff}}}{m} = \sqrt{\frac{\Lambda_{\text{Landau}}}{m}} \gg 1 \quad \Rightarrow \quad m \ll \Lambda_{\text{cutoff}},$$

and

$$\frac{\Lambda_{\text{cutoff}}}{\Lambda_{\text{Landau}}} = \sqrt{\frac{m}{\Lambda_{\text{Landau}}}} \ll 1 \quad \Rightarrow \quad \Lambda_{\text{cutoff}} \ll \Lambda_{\text{Landau}}.$$

Hence

$$\boxed{m \ll \Lambda_{\text{cutoff}} \ll \Lambda_{\text{Landau}} \quad \text{when} \quad \Lambda_{\text{Landau}} \gg m.}$$

This choice places the EFT cutoff parametrically below the Landau pole (so the coupling remains perturbative and unknown UV strong-coupling effects are excluded), while keeping it parametrically above the light scale m (so there is a wide window $m \ll E \ll \Lambda_{\text{cutoff}}$ where the EFT is valid and predictive). $\square$

5. Power-suppressed errors vs perturbative errors. Assume you compute an observable $\mathcal{O}$ to N loops, so the perturbative truncation error scales like $\alpha(\mu)^{N+1}$. Meanwhile EFT cutoff effects scale like $(E/\Lambda_{\text{cutoff}})^p$ for some $p > 0$. Explain how choosing Λ_{cutoff} parametrically below Λ_{Landau} can make cutoff errors smaller than the perturbative truncation error at fixed N .

6. Running-coupling estimate of EFT validity. Take $E \ll \Lambda_{\text{cutoff}}$. Using the one-loop running, estimate $\alpha(E)$ and discuss why the EFT can remain weakly coupled even though $\alpha(\mu)$ blows up at Λ_{Landau} .

7. Interpretation: “asymptotic series plus EFT.” Summarize the logic: (i) perturbation theory is an asymptotic expansion, (ii) EFT adds a controlled expansion in $E/\Lambda_{\text{cutoff}}$, and (iii) the Landau pole does not obstruct predictivity at energies $E \ll \Lambda_{\text{cutoff}}$.

Solution Manual of Problem Set 33

Bufan Zheng

bufan@ecc.u-tokyo.ac.jp

Gauge field in modern convention: $\mathcal{L} \supset -\frac{1}{4e^2} F^2$. Gauge transformation $\delta A_\mu = -\partial_\mu \alpha$.

1. Lorenz gauge fixing term. Add $\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi e^2} (\partial \cdot A)^2$. Derive the quadratic operator for A_μ in momentum space and write the propagator $\langle A_\mu A_\nu \rangle(p)$.

Solution. (33.1) In problem 23.4 we derived a generalized result of propagator for massive vector fields in R_ξ . So here we just quote the result:

$$D_{\mu\nu}(p) = -ie^2 \left[\frac{\eta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}}{p^2 - m_A^2} + \frac{\xi \frac{p_\mu p_\nu}{p^2}}{p^2 - \xi m_A^2} \right]$$

Now, sending m_A to 0 and add the regulator by hand, **MY FINAL ANSWER IS:**

$$D_{\mu\nu}(p) = \frac{ie^2}{p^2 + i\epsilon} \left(\eta_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2 + i\epsilon} \right)$$

□

2. FP determinant in abelian Lorenz gauge. For gauge condition $G(A) = \partial \cdot A$, compute $\delta G / \delta \alpha$. Show the FP determinant is $\det(\partial^2)$ and hence the ghost action is free:

$$S_{\text{gh}} = \int d^D x \, b (-\partial^2) c.$$

Solution. (33.2) For the abelian gauge condition $G(A) = \partial \cdot A$ and gauge variation $\delta_\alpha A_\mu = -\partial_\mu \alpha$,

$$\delta_\alpha G = \partial^\mu \delta_\alpha A_\mu = -\partial^\mu \partial_\mu \alpha = -\partial^2 \alpha, \quad \Rightarrow \quad \frac{\delta G}{\delta \alpha} = -\partial^2.$$

Hence the FP determinant is

$$\Delta_{\text{FP}} = \det(-\partial^2),$$

Recall that for grassmannian number, we have:

$$\int \prod_{i=1}^n d\bar{\psi}_i d\psi_i \exp \left(- \sum_{i,j=1}^n \bar{\psi}_i A_{ij} \psi_j \right) = \det A.$$

So the FP determinant can be exponentiated by anticommuting ghosts b, c to give a free ghost action,

$$S_{\text{gh}} = \int d^D x \, b (-\partial^2) c$$

□

3. Ghost decoupling in linear gauges. Show explicitly that in abelian theory the ghost fields do not couple to A in any linear gauge $G(A) = L^\mu A_\mu$. Identify the operator whose determinant appears.

Solution. (33.3) In an abelian theory with a linear gauge condition $G(A) = L^\mu A_\mu$, where L^μ is a fixed (field-independent) linear differential operator, we only need to mimic the solution to the previous problem:

$$\delta_\alpha G = L^\mu \delta_\alpha A_\mu = -L^\mu \partial_\mu \alpha, \quad \Rightarrow \quad \frac{\delta G}{\delta \alpha} = -(L^\mu \partial_\mu).$$

Therefore the FP determinant is

$$\Delta_{\text{FP}} = \det(-(L^\mu \partial_\mu)),$$

and the ghost action is

$$S_{\text{gh}} = \int d^D x \, b (-L^\mu \partial_\mu) c.$$

Since $L^\mu \partial_\mu$ does not depend on A_μ , the ghosts do not couple to A_μ and hence decouple. Therefore, in QED, usually people just ignore ghost field, only the gauge fixing term left. $\square$

4. BRST transformations (abelian). Introduce the ghost c , antighost b , and Nakanishi–Lautrup field B . Define BRST variations

$$Q_{\text{BRST}} A_\mu = -\partial_\mu c, \quad Q_{\text{BRST}} c = 0, \quad Q_{\text{BRST}} b = B, \quad Q_{\text{BRST}} B = 0.$$

Verify $Q_{\text{BRST}}^2 = 0$ on A_μ, c, b, B . After eliminating B using its algebraic equation of motion from

$$\mathcal{L}_{\text{gf}} = \frac{1}{e^2} \left(\frac{\xi}{2} B^2 - B \partial \cdot A \right),$$

show $B = \frac{1}{\xi} \partial \cdot A$ and hence (on-shell in B) $Q_{\text{BRST}} b = \frac{1}{\xi} \partial \cdot A$.

Solution. (33.4) Nilpotency follows immediately:

$$Q_{\text{BRST}}^2 A_\mu = -\partial_\mu (Q_{\text{BRST}} c) = 0, \quad Q_{\text{BRST}}^2 c = 0, \quad Q_{\text{BRST}}^2 b = Q_{\text{BRST}} B = 0, \quad Q_{\text{BRST}}^2 B = 0.$$

With

$$\mathcal{L}_{\text{gf}} = \frac{1}{e^2} \left(\frac{\xi}{2} B^2 - B \partial \cdot A \right),$$

the algebraic equation of motion for B is

$$\frac{\partial \mathcal{L}_{\text{gf}}}{\partial B} = \frac{1}{e^2} (\xi B - \partial \cdot A) = 0 \quad \Rightarrow \quad B = \frac{1}{\xi} \partial \cdot A.$$

Therefore, on-shell in B ,

$$Q_{\text{BRST}} b = B = \frac{1}{\xi} \partial \cdot A.$$

$\square$

5. Abelian Higgs in R_ξ : ghost mass. In the abelian Higgs model, take gauge fixing

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} (\partial \cdot A - \xi m_A \pi)^2, \quad m_A \equiv ev.$$

Compute $\delta G / \delta \alpha$ and show the ghosts acquire a mass

$$m_{\text{gh}}^2 = \xi m_A^2.$$

Solution. (33.5) In the abelian Higgs model, choose the R_ξ gauge condition

$$G(A, \pi) = \partial \cdot A - \xi m_A \pi, \quad \mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} G^2, \quad m_A \equiv ev.$$

Using $\delta_\alpha A_\mu = -\partial_\mu \alpha$ and $\delta_\alpha \pi = m_A \alpha$ (since $\delta\phi = ie\alpha$ and $ev = m_A$), one finds

$$\delta_\alpha G = \partial^\mu \delta_\alpha A_\mu - \xi m_A \delta_\alpha \pi = -\partial^2 \alpha - \xi m_A^2 \alpha = -(\partial^2 + \xi m_A^2) \alpha,$$

so

$$\frac{\delta G}{\delta \alpha} = -(\partial^2 + \xi m_A^2), \quad \Delta_{\text{FP}} = \det(\partial^2 + \xi m_A^2).$$

Exponentiating the determinant gives

$$S_{\text{gh}} = \int d^D x \, b \, (-\partial^2 - \xi m_A^2) \, c,$$

so the ghosts acquire a mass

$$\boxed{m_{\text{gh}}^2 = \xi m_A^2}.$$

□

6. Unitary gauge FP determinant. Unitary gauge is $\pi = 0$. Under an infinitesimal gauge transformation $\delta\pi = v\alpha + \mathcal{O}(\pi)$. Compute the FP determinant at $\pi = 0$ and show it is field-independent at leading order.

7. Renormalizability signal (power counting). In 4D abelian Higgs, compare propagator behavior in unitary gauge versus R_ξ gauge at large momentum. Show the R_ξ gauge propagator falls as $1/p^2$ in all components, supporting power-counting renormalizability.

Solution Manual of Problem Set 34

Bufan Zheng

bufan@ecc.u-tokyo.ac.jp

Let $G = SU(N)$ with generators T^a in the fundamental, normalized by $\text{tr}(T^a T^b) = \frac{1}{2} \delta^{ab}$. Structure constants $[T^a, T^b] = i f^{abc} T^c$.

1. Structure constants identity. Show that f^{abc} are totally antisymmetric under the above normalization and compute $f^{abc} f^{abd}$ in terms of N (i.e. C_A).

Solution. (34.1) From our convention:

$$f^{abc} = -2i \text{tr}([T^a, T^b] T^c)$$

so $f^{abc} = -f^{bac}$. Using cyclicity and invariance of the trace,

$$\text{tr}([T^a, T^b] T^c) = \text{tr}(T^a [T^b, T^c]),$$

which implies total antisymmetry in (a, b, c) . For $SU(N)$ group, we have the following completeness relation:

$$(T^c)_{ij} (T^c)_{kl} = \frac{1}{2} \left(\delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl} \right) \implies \sum_c T^c X T^c = \frac{1}{2} \text{tr}(X) \mathbf{1} - \frac{1}{2N} X, \quad \sum_c T^c T^c = \frac{N^2 - 1}{2N} \mathbf{1}$$

which implies:

$$\begin{aligned} f^{abc} f^{abd} &= f^{cab} f^{dab} = -2 \sum_a \text{tr}([T^c, T^a][T^d, T^a]) \\ &= -2 \sum_a \text{tr}((T^c T^a - T^a T^c)(T^d T^a - T^a T^d)) \\ &= -2 \left(\text{tr}(T^c (\sum_a T^a T^d T^a)) - \text{tr}(T^c (\sum_a T^a T^a) T^d) - \text{tr}(\sum_a T^a (T^c T^d) T^a) + \text{tr}((\sum_a T^a T^c T^a) T^d) \right) \\ &= -2 \left(\text{tr}(T^c (-\frac{1}{2N} T^d)) - \text{tr}(T^c (\frac{N^2-1}{2N} \mathbf{1}) T^d) - \text{tr}(\frac{1}{2} \text{tr}(T^c T^d) \mathbf{1}) - \frac{1}{2N} \text{tr}(T^c T^d) + \text{tr}((-\frac{1}{2N} T^c) T^d) \right) \\ &= -2 \left(-\frac{1}{2N} \text{tr}(T^c T^d) - \frac{N^2-1}{2N} \text{tr}(T^c T^d) - \frac{1}{2} \text{tr}(T^c T^d) \text{tr}(\mathbf{1}) + \frac{1}{2N} \text{tr}(T^c T^d) - \frac{1}{2N} \text{tr}(T^c T^d) \right) \\ &= -2 \left(-\frac{N^2-1}{2N} \text{tr}(T^c T^d) - \frac{1}{2} \text{tr}(T^c T^d) N \right) = -2 \left(-\frac{N^2}{2} \text{tr}(T^c T^d) \right) = \boxed{N \delta^{cd}}, \end{aligned}$$

□

2. Casimirs. For $SU(N)$, show $T^a T^a = C_F \mathbf{1}$ in the fundamental and compute $C_F = (N^2 - 1)/(2N)$. Also state $C_A = N$ for the adjoint.

Solution. (34.2) The generator in the adjoint representation can be written as:

$$T_{Abc}^a = i f^{abc}$$

Which implies:

$$T_A^a T_A^a = -f^{acb} f^{abd} = f^{abc} f^{abd} = C_A \delta^{cd}$$

In the last problem, we have computed $C_A = N$. To compute C_F , we can use the completeness relation:

$$(T^c)_{ij} (T^c)_{jk} = \frac{1}{2} \left(\delta_{ik} \delta_{jj} - \frac{1}{N} \delta_{ij} \delta_{jk} \right) = \frac{1}{2} \left(N - \frac{1}{N} \right) \delta_{ik} = \frac{N^2 - 1}{2N} \delta_{ik}$$

Hence

$$C_F = \frac{N^2 - 1}{2N}$$

□

3. Classical nonabelian Noether current. For a complex scalar ϕ in the fundamental with Lagrangian $\mathcal{L} = |\partial_\mu \phi|^2 - V(|\phi|^2)$, under global $SU(N)$: $\delta\phi = i\epsilon^a T^a \phi$, compute the Noether current $j^{a\mu}$.

Solution. (34.3) The Noether current is

$$j^{a\mu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \delta\phi^* = \partial^\mu \phi^* (i\epsilon^a T^a \phi) + \partial^\mu \phi (-i\epsilon^a T^a \phi^*).$$

Removing the common factor ϵ^a gives

$$j^{a\mu} = i \left(\partial^\mu \phi^* T^a \phi - \phi^* T^a \partial^\mu \phi \right).$$

□

4. Nonabelian covariant derivative and field strength. Define $A_\mu = A_\mu^a T^a$ and $D_\mu = \partial_\mu - iA_\mu$. Show $[D_\mu, D_\nu] = -iF_{\mu\nu}$ and derive

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu].$$

Solution. (33.4) For any test field Ψ one finds

$$[D_\mu, D_\nu]\Psi = (\partial_\mu - iA_\mu)(\partial_\nu - iA_\nu)\Psi - (\mu \leftrightarrow \nu) = -i(\partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu])\Psi.$$

Hence

$$[D_\mu, D_\nu] = -iF_{\mu\nu},$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu].$$

□

5. One-loop β -function sign in 4D. In 4D, pure Yang–Mills has $\beta(g) = -\frac{11}{3} \frac{C_A}{16\pi^2} g^3$. Show this implies asymptotic freedom. Solve for $g^2(\mu)$ given $g(\mu_0)$.

Solution. (34.5) Starting from

$$\mu \frac{dg}{d\mu} = -\frac{11}{3} \frac{C_A}{16\pi^2} g^3,$$

the right-hand side is negative for $g > 0$, so $g(\mu)$ decreases as μ increases which means the pure YM theory is asymptotic freedom. Separating variables,

$$\frac{dg}{g^3} = -\frac{11C_A}{48\pi^2} d \ln \mu.$$

Integrating from μ_0 to μ ,

$$-\frac{1}{2} \left(\frac{1}{g^2(\mu)} - \frac{1}{g^2(\mu_0)} \right) = -\frac{11C_A}{48\pi^2} \ln \frac{\mu}{\mu_0},$$

so **MY FINAL ANSWER IS**

$$\frac{1}{g^2(\mu)} = \frac{1}{g^2(\mu_0)} + \frac{11C_A}{24\pi^2} \ln \frac{\mu}{\mu_0}.$$

□

6. Add matter in representations. In 4D, include n_f Dirac fermions and n_s complex scalars in representation R with index $T(R)$ defined by $\text{tr}_R(T^a T^b) = T(R)\delta^{ab}$. Write the one-loop coefficient b_0 in

$$\beta(g) = -\frac{b_0}{16\pi^2}g^3$$

and specialize to $R = \text{fundamental of } SU(N)$ with $T(F) = 1/2$.

Solution. (34.6) The one-loop beta function can be written as

$$\beta(g) = -\frac{b_0}{16\pi^2}g^3, \quad b_0 = \frac{11}{3}C_A - \frac{4}{3}n_f T(R) - \frac{1}{6}n_s T(R).$$

Specializing to $R = F$, i.e. the fundamental of $SU(N)$, with $T(F) = \frac{1}{2}$ and $C_A = N$, one finds

$$b_0 = \frac{11}{3}N - \frac{2}{3}n_f - \frac{1}{12}n_s.$$

□

7. Anomaly cancellation and mass terms (nonabelian). In 4D, the $SU(N)^3$ anomaly from left-handed Weyl fermions in reps R_i is proportional to $\sum_i A(R_i)$ where $A(R)$ is the cubic index. Explain quantitatively why a vectorlike pair $R \oplus \bar{R}$ makes zero net contribution and can admit a gauge-invariant mass.

Solution Manual of Problem Set 35

Bufan Zheng

bufan@g.ecc.u-tokyo.ac.jp

Let $A_\mu = A_\mu^a T^a$ for $SU(N)$ with $[T^a, T^b] = if^{abc}T^c$. For an infinitesimal gauge parameter $\alpha = \alpha^a T^a$, take

$$\delta A_\mu = \partial_\mu \alpha + i[A_\mu, \alpha],$$

which in components is $\delta A_\mu^a = \partial_\mu \alpha^a + f^{abc} A_\mu^b \alpha^c$.

1. Variation of Lorenz gauge condition. Take $G^a = \partial^\mu A_\mu^a$. Compute δG^a under $\delta A_\mu^a = \partial_\mu \alpha^a + f^{abc} A_\mu^b \alpha^c$. Write $\delta G^a = M^{ab} \alpha^b$ and identify M^{ab} .

Solution. (35.1)

$$\begin{aligned} \delta G^a &= \partial^\mu \delta A_\mu^a = \partial^\mu (\partial_\mu \alpha^a + f^{abc} A_\mu^b \alpha^c) \\ &= \square \alpha^a + f^{abc} \partial^\mu (A_\mu^b \alpha^c) \\ &= \square \alpha^a + f^{abc} (\partial^\mu A_\mu^b) \alpha^c + f^{abc} A_\mu^b \partial^\mu \alpha^c. \end{aligned}$$

MY FINAL ANSWER IS

$$M^{ab}(x, y) = \frac{\delta G^a(x)}{\alpha^b(y)} = \partial^\mu D_\mu^{ab} \delta(x - y), \quad D_\mu^{ab} = \delta^{ab} \partial_\mu + f^{acb} A_\mu^c$$

□

2. FP ghost action. Show that the FP determinant $\det M$ can be represented by ghosts c^a and antighosts b^a with action

$$S_{\text{gh}} = \int d^D x b^a M^{ab} c^b.$$

Write S_{gh} explicitly and identify the ghost–gauge interaction term.

Solution. (35.2) Recall the Gaussian integral of Grassmannian matrices:

$$\det M \propto \int \mathcal{D}c \mathcal{D}b \exp \left(\int d^D x b^a M^{ab} c^b \right),$$

so that

$$S_{\text{gh}} = \int d^D x b^a M^{ab} c^b = \int d^D x b^a \partial^\mu (D_\mu c)^a.$$

Using $(D_\mu c)^a = \partial_\mu c^a + f^{abc} A_\mu^b c^c$, one finds

$$\begin{aligned} S_{\text{gh}} &= \int d^D x b^a \partial^\mu (\partial_\mu c^a + f^{abc} A_\mu^b c^c) \\ &= \int d^D x \left[b^a \square c^a + f^{abc} b^a \partial^\mu (A_\mu^b c^c) \right] \\ &= \int d^D x \left[b^a \square c^a + f^{abc} b^a (\partial^\mu A_\mu^b) c^c + f^{abc} b^a A_\mu^b \partial^\mu c^c \right]. \end{aligned}$$

By integral by part, we can rewrite it as

$$S_{\text{gh}} = \int d^D x \left[-\partial^\mu b^a \partial_\mu c^a + g f^{abc} A_\mu^c \partial^\mu b^a c^b \right]$$

up to a total derivative.

□

3. Ghost coupling is unavoidable. In any covariant ξ -gauge with $G^a = \partial \cdot A^a$, show that M^{ab} contains a term linear in A and thus ghosts couple to A whenever $f^{abc} \neq 0$.

Solution. (35.3) In the previous Problem 1 we have already computed M^{ab} , and after dropping the annoying $\delta(x - y)$, it is not hard to see that

$$M^{ab} = \square \delta^{ab} + f^{acb}(\partial^\mu A_\mu^c) + f^{acb}A_\mu^c \partial^\mu$$

The last term is linear in A whenever $f^{abc} \neq 0$. Hence the ghost action $S_{\text{gh}} = \int d^D x b^a M^{ab} c^b$ necessarily contains a term linear in A , for example $-f^{abc}(\partial^\mu b^a)A_\mu^b c^c$, so ghosts couple to the gauge field in any such nonabelian gauge fixing. $\square$

4. BRST transformations. Introduce the ghost c^a , antighost b^a , and Nakanishi–Lautrup field B^a . Define BRST:

$$Q_{\text{BRST}} A_\mu^a = (D_\mu c)^a, \quad Q_{\text{BRST}} c^a = -\frac{1}{2} f^{abc} c^b c^c, \quad Q_{\text{BRST}} b^a = B^a, \quad Q_{\text{BRST}} B^a = 0,$$

with $(D_\mu c)^a = \partial_\mu c^a + f^{abc} A_\mu^b c^c$. Verify $Q_{\text{BRST}}^2 A_\mu^a = 0$ using the Jacobi identity.

Solution. (35.4)

$$\begin{aligned} Q_{\text{BRST}}^2 A_\mu^a &= Q_{\text{BRST}}(D_\mu c)^a \\ &= \partial_\mu(Q_{\text{BRST}} c^a) + f^{abc}(Q_{\text{BRST}} A_\mu^b) c^c + f^{abc} A_\mu^b (Q_{\text{BRST}} c^c) \\ &= -\frac{1}{2} f^{ade} \partial_\mu(c^d c^e) + f^{abc}(D_\mu c)^b c^c - \frac{1}{2} f^{abc} A_\mu^b f^{cde} c^d c^e \\ &= -\frac{1}{2} f^{ade} \partial_\mu(c^d c^e) + f^{abc}(D_\mu c)^b c^c - \frac{1}{2} f^{abc} A_\mu^b f^{cde} c^d c^e \\ &= -\frac{1}{2} f^{ade} \left[(\partial_\mu c^d) c^e + c^d (\partial_\mu c^e) \right] + f^{abc} \left[\partial_\mu c^b + f^{bde} A_\mu^d c^e \right] c^c - \frac{1}{2} f^{abc} A_\mu^b f^{cde} c^d c^e \\ &= \left(-\frac{1}{2} f^{ade} (\partial_\mu c^d) c^e - \frac{1}{2} f^{ade} c^d (\partial_\mu c^e) + f^{abc} (\partial_\mu c^b) c^c \right) + f^{abc} f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{abc} f^{cde} A_\mu^b c^d c^e \\ &= f^{abc} f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{abc} f^{cde} A_\mu^b c^d c^e \\ &= -\frac{1}{2} \left(f^{abm} f^{mcd} + f^{acm} f^{mdb} + f^{adm} f^{mbc} \right) A_\mu^b c^c c^d \\ &= 0 \end{aligned}$$

In the last step, we used the Jacobi identity. $\square$

5. Gauge-fixing as BRST-exact. Take $G^a = \partial \cdot A^a$. Show that the gauge-fixing plus ghost Lagrangian can be written (up to a total derivative) as

$$\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}} = Q_{\text{BRST}} \left[b^a \left(G^a - \frac{\xi}{2} B^a \right) \right].$$

Work out the result explicitly and verify it contains: (i) a term $B^a G^a$, (ii) a term $-\frac{\xi}{2} B^a B^a$, and (iii) the ghost term $b^a M^{ab} c^b$. (Optionally eliminate B^a to obtain $\mathcal{L}_{\text{gf}} = -(2\xi)^{-1} (G^a)^2$.)

Solution. (35.5)

$$\begin{aligned}\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}} &= (Q_{\text{BRST}} b^a) \left(G^a - \frac{\xi}{2} B^a \right) + b^a \left(Q_{\text{BRST}} G^a - \frac{\xi}{2} Q_{\text{BRST}} B^a \right) \\ &= B^a G^a - \frac{\xi}{2} B^a B^a + b^a Q_{\text{BRST}} G^a.\end{aligned}$$

Since $Q_{\text{BRST}} G^a = \partial^\mu (Q_{\text{BRST}} A_\mu^a) = \partial^\mu (D_\mu c)^a \equiv M^{ab} c^b$, one gets

$$\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}} = B^a G^a - \frac{\xi}{2} B^a B^a + b^a M^{ab} c^b,$$

which contains (i) $B^a G^a$, (ii) $-\frac{\xi}{2} B^a B^a$, (iii) the ghost term $b^a M^{ab} c^b$. Integrating out B^a gives

$$\frac{\delta}{\delta B^a} (\mathcal{L}_{\text{gh}} + \mathcal{L}_{\text{gf}}) = G^a - \xi B^a = 0 \Rightarrow B^a = \frac{1}{\xi} G^a$$

plugging into $\mathcal{L}_{\text{gf}}$ gives:

$$\mathcal{L}_{\text{gf}} = \frac{1}{\xi} G^a G^a - \frac{1}{2\xi} G^a G^a = -\frac{1}{2\xi} (G^a)^2$$

Which is the usual form we add in YM lagrangian as a gauge fixing term in ξ -gauge. □

6. Propagators in ξ -gauge (free part). Expand Yang–Mills around $A = 0$. Write the momentum-space propagators for A_μ^a and for ghosts c^a, b^a including ξ dependence.

7. Check: abelian limit. Set $f^{abc} = 0$. Show the ghost–gauge interaction vanishes and the ghost determinant reduces to $\det(\partial^2)$. Compare with the abelian FP procedure.

Solution. (35.7) Set $f^{abc} = 0$. Then the gauge variation becomes $\delta A_\mu^a = \partial_\mu \alpha^a$ and

$$M^{ab} = \partial^\mu \partial_\mu \delta^{ab} = \partial^2 \delta^{ab},$$

which implies the FP determinant reduces to:

$$\Delta_{\text{FP}} = \det M^{ab} = \det(\partial^2)^{\# \text{ of color}} \det(\delta^{ab}) = \det(\partial^2)^{\# \text{ of color}}$$

Therefore the ghost action reduces to a free quadratic form,

$$S_{\text{gh}} = \int d^D x b^a \partial^2 c^a,$$

and there is no ghost–gauge interaction term. This matches our previous result in Problem Set 33. □